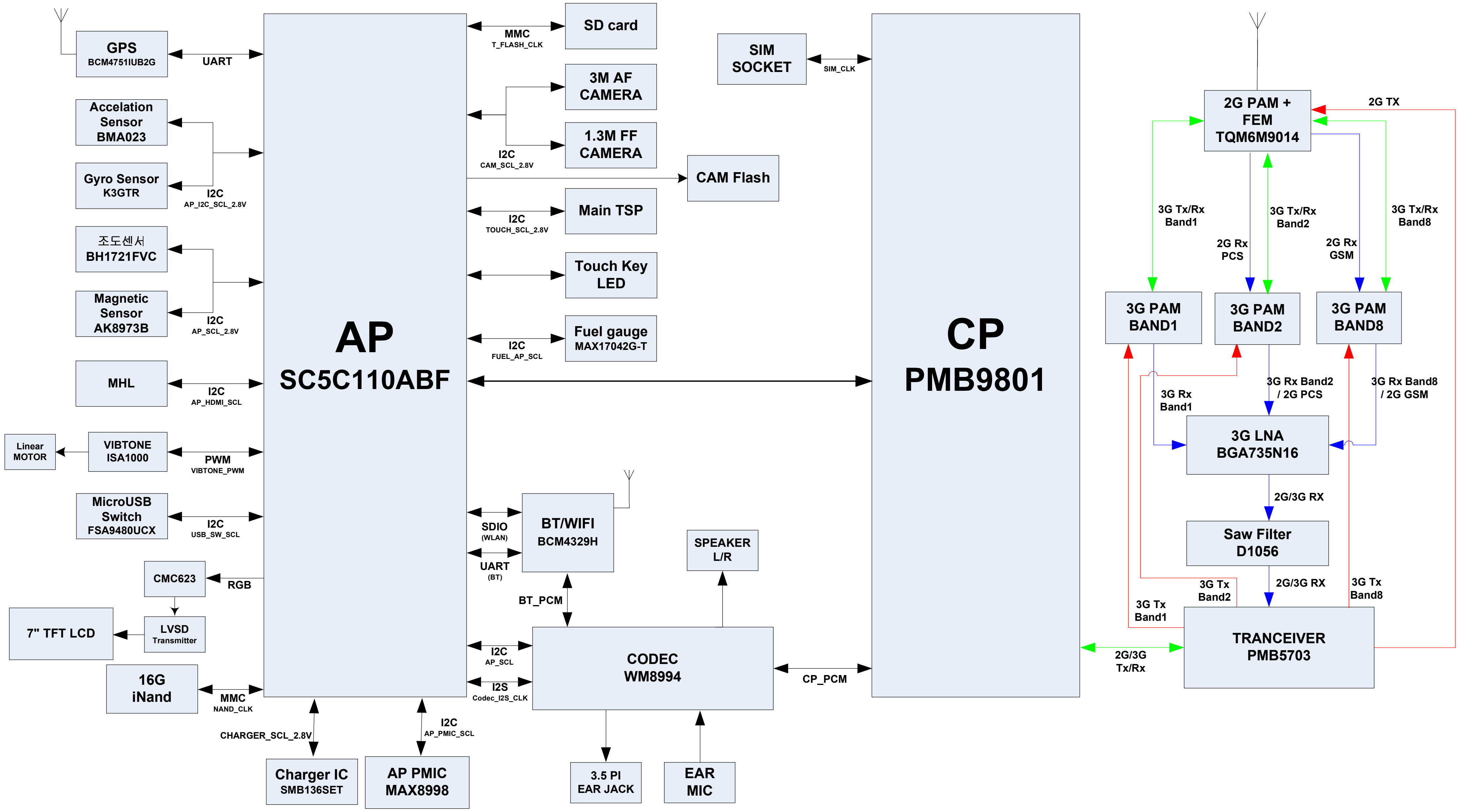
8. 3 级维修

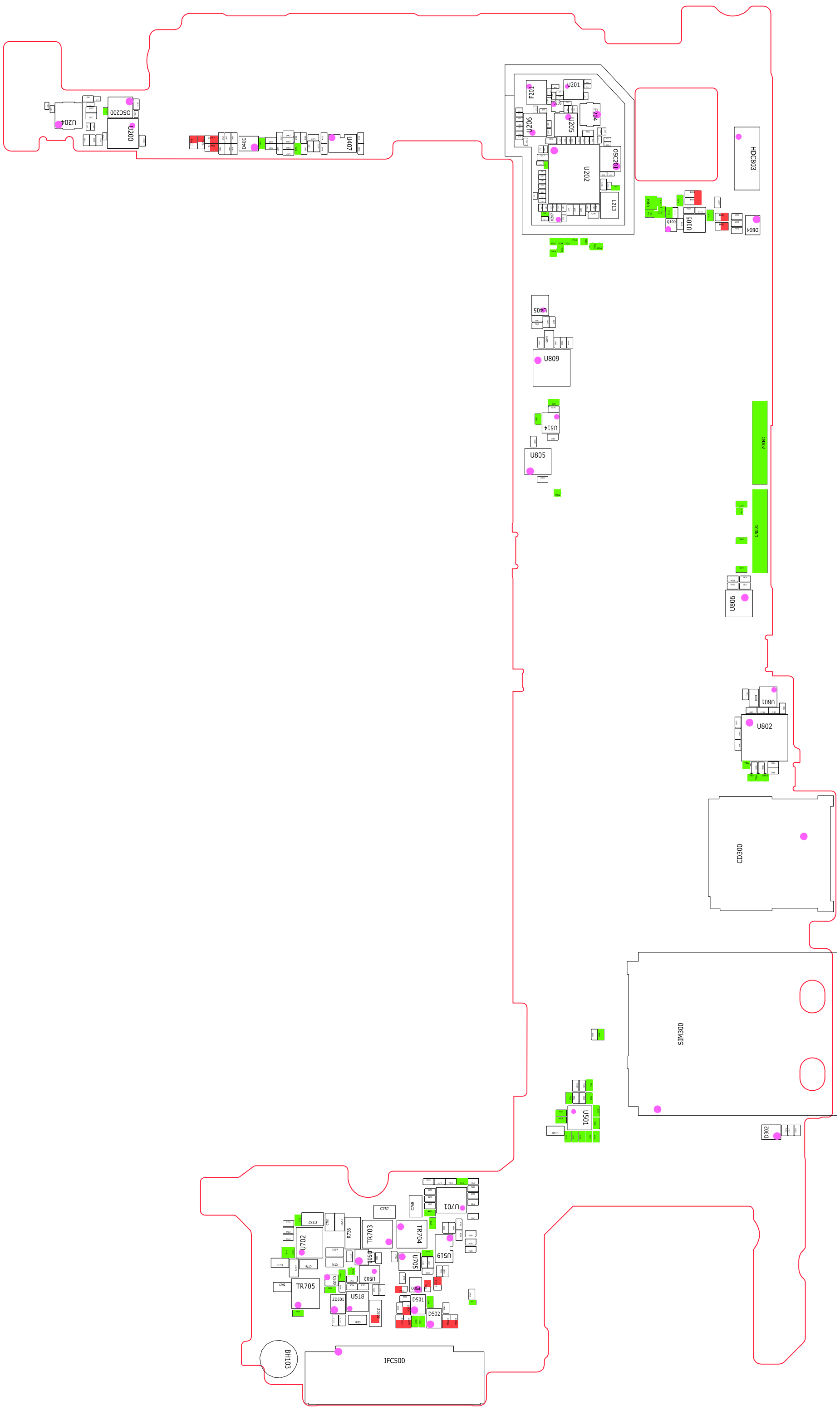
8-1.框图



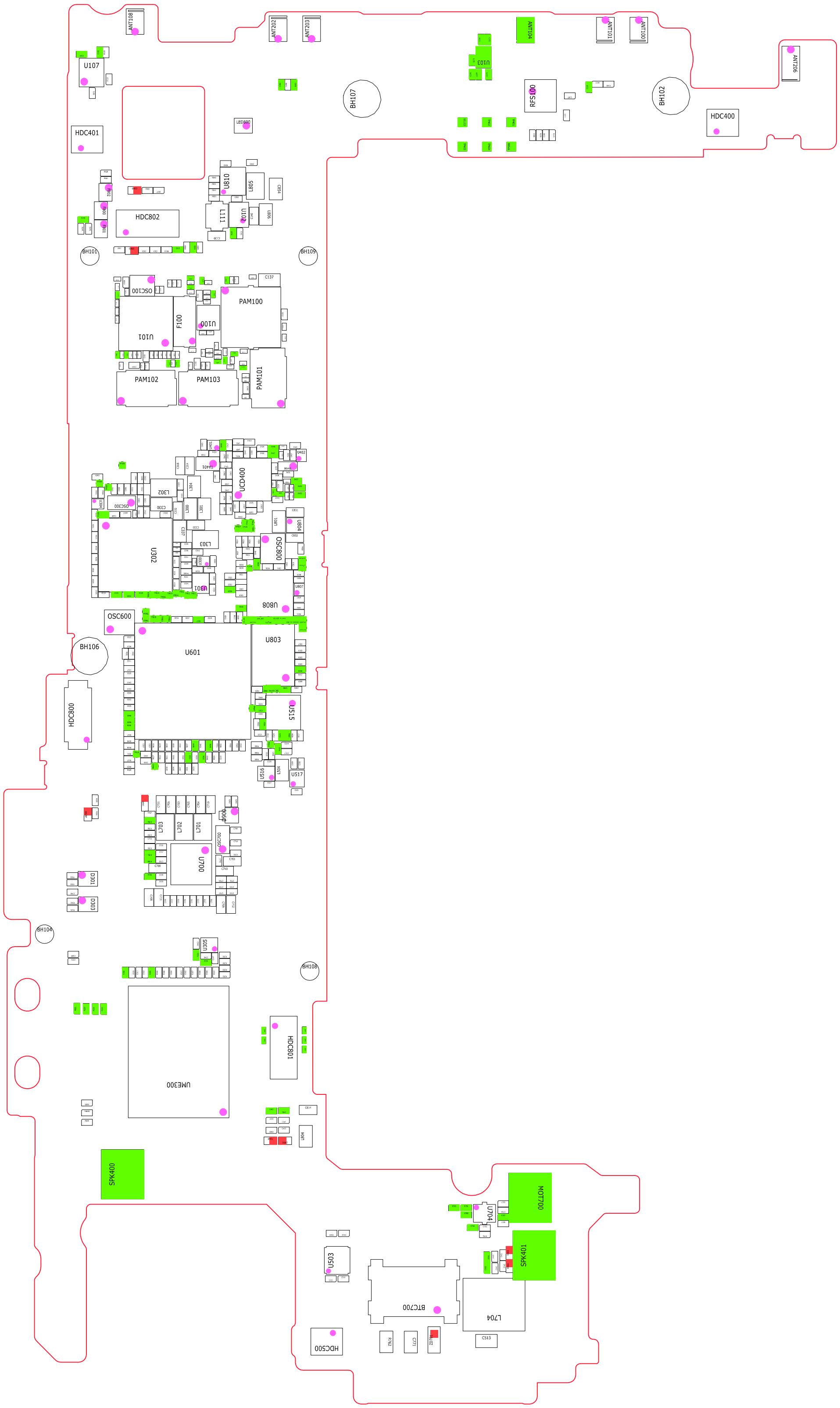
GPS
BCM4751IUB2G
UART
Accelation Sensor BMA023
Gyro Sensor K3GTR
I2C
AP_I2C_SCL_2.8V
조도센서 BH1721FVC
Magnetic Sensor AK8973B
I2C
AP_SCL_2.8V
MHL
I2C
AP_HDMI_SCL
Linear MOTOR
VIBTONE ISA1000
PWM
VIBTONE_PWM
MicroUSB Switch FSA9480UCX
I2C
USB_SW_SCL
CMC623
RGB
7" TFT LCD
LVSD Transmitter
16G iNand
MMC
NAND_CLK
AP
SC5C110ABF
CHARGER_SCL_2.8V
Charger IC SMB136SET
I2C
AP_PMIC_SCL
AP PMIC MAX8998
MMC
T_FLASH_CLK
SD card
3M AF CAMERA
1.3M FF CAMERA
I2C
CAM_SCL_2.8V
CAM Flash
I2C
TOUCH_SCL_2.8V
Main TSP
Touch Key LED
I2C
FUEL_AP_SCL
Fuel gauge MAX17042G-T
SDIO (WLAN)
UART (BT)
BT/WIFI BCM4329H
BT_PCM
I2C
AP_SCL
I2S
Codec_I2S_CLK
CODEC WM8994
SPEAKER L/R
3.5 PI EAR JACK
EAR MIC
CP_PCM
SIM SOCKET
SIM_CLK
CP
PMB9801
2G PAM + FEM TQM6M9014
2G TX
3G Tx/Rx Band1
3G Tx/Rx Band2
3G Tx/Rx Band8
2G Rx PCS
2G Rx GSM
3G PAM BAND1
3G PAM BAND2
3G PAM BAND8
3G Rx Band1
3G Rx Band2 / 2G PCS
3G Rx Band8 / 2G GSM
3G LNA BGA735N16
2G/3G RX
Saw Filter D1056
2G/3G RX
3G Tx Band2
3G Tx Band8
3G Tx Band1
2G/3G Tx/Rx
TRANCEIVER PMB5703

8-2.主板图

8-2-1.顶部

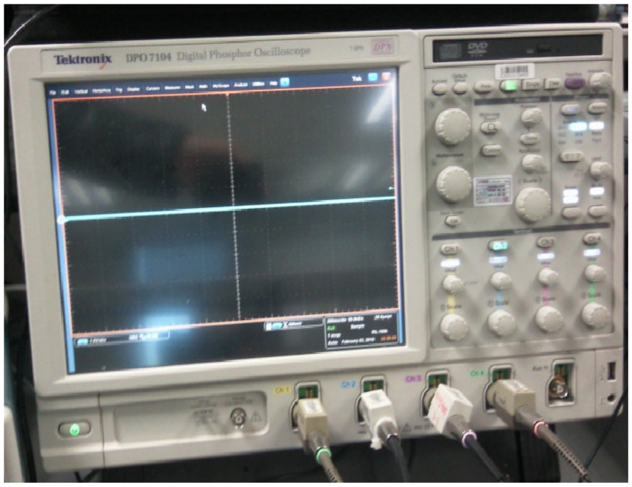


8-2-2.底部



8-3.故障排除流程图

设备



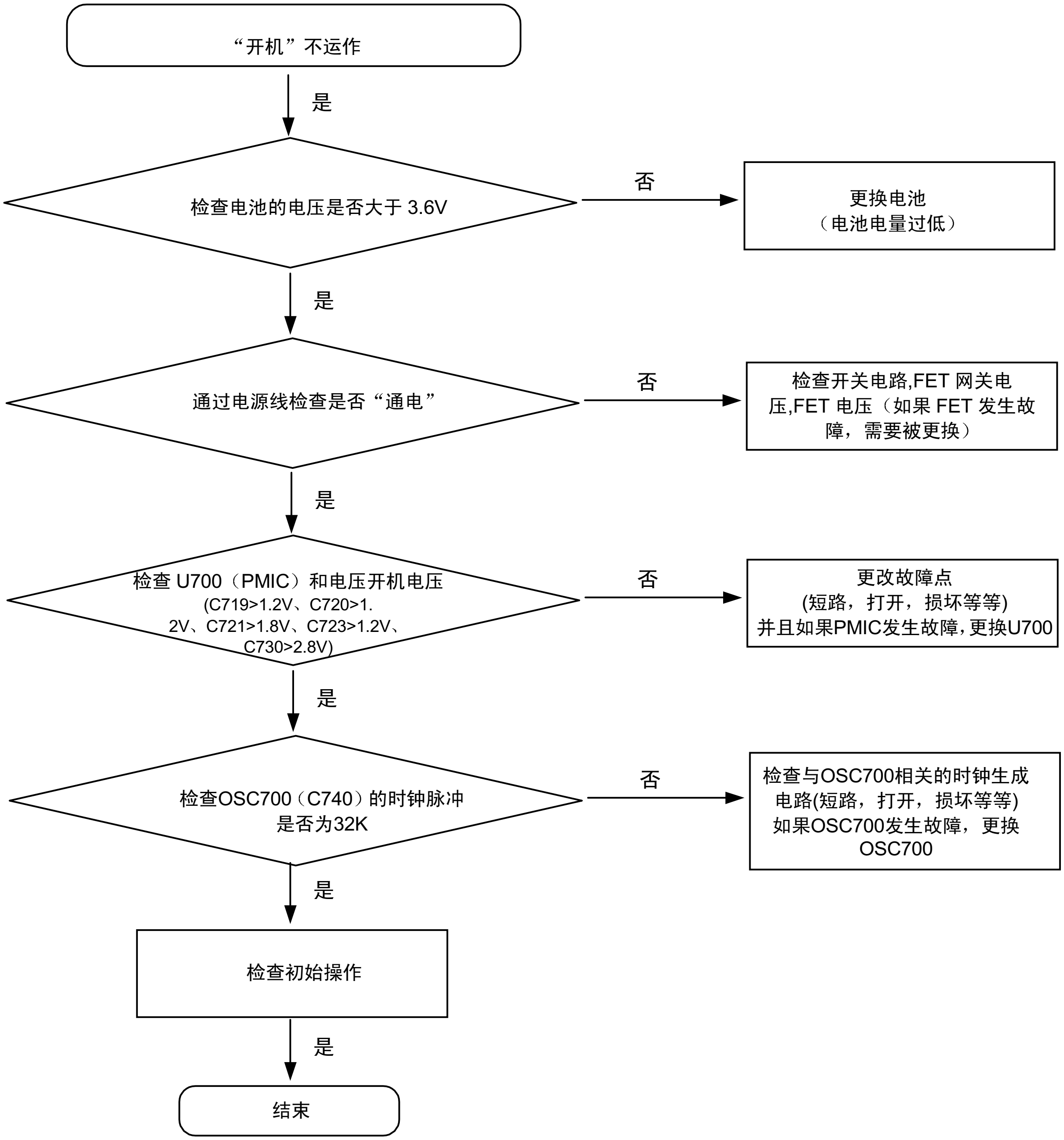
┐ 示波器

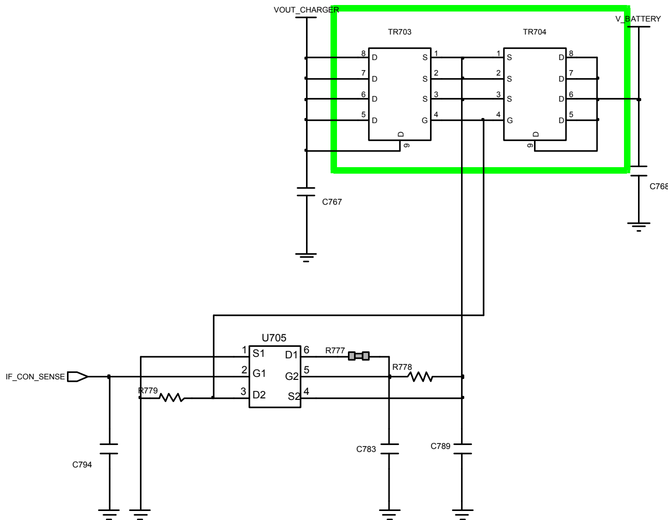
┐ 数字万用表

┐ 电源供电器

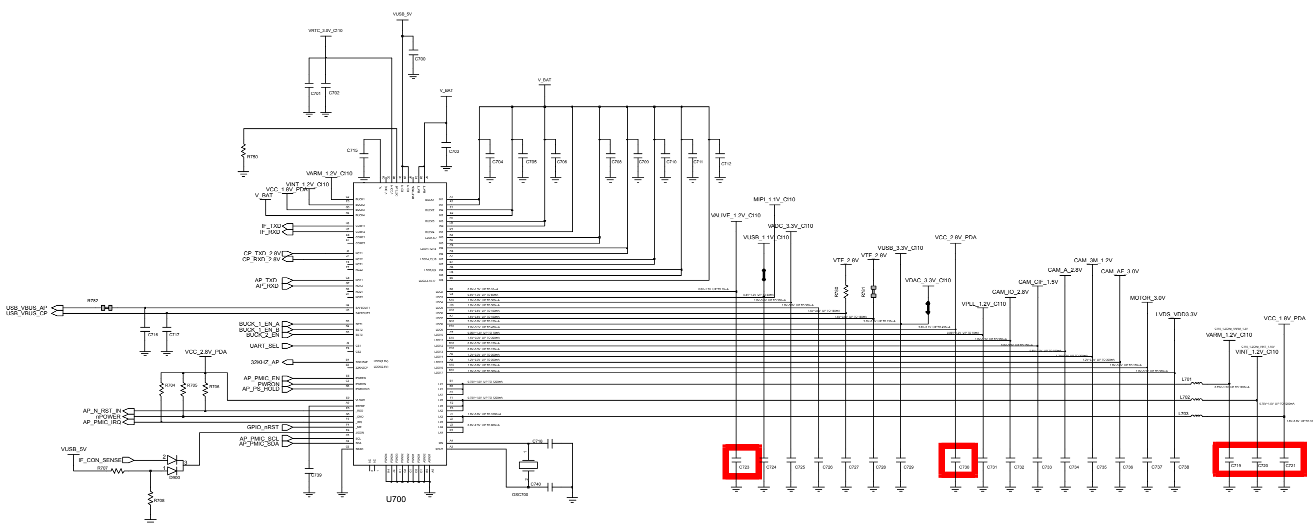
┐ +字螺丝刀，小镊子

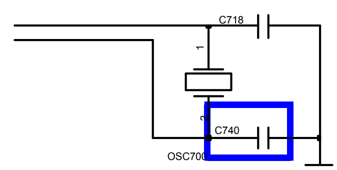
8-3-1开机


“开机”不运作
是
检查电池的电压是否大于 3.6V
否
更换电池
（电池电量过低）
是
通过电源线检查是否“通电”
否
检查开关电路,FET 网关电压,FET 电压（如果 FET 发生故障，需要被更换）
是
检查 U700（PMIC）和电压开机电压
(C719>1.2V、C720>1.2V、C721>1.8V、C723>1.2V、C730>2.8V)
否
更改故障点
(短路，打开，损坏等等)
并且如果PMIC发生故障,更换U700
是
检查OSC700（C740）的时钟脉冲是否为32K
否
检查与OSC700相关的时钟生成电路(短路，打开，损坏等等)
如果OSC700发生故障，更换OSC700
是
检查初始操作
是
结束

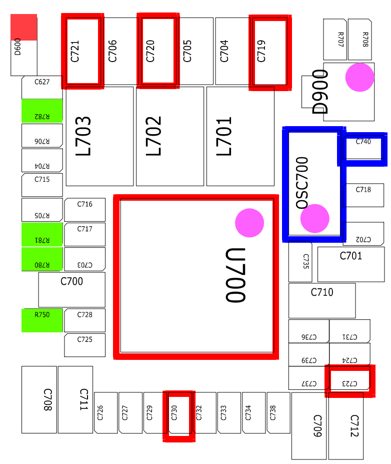

VOUT_CHARGER
V_BATTERY
TR703
TR704
C767
C768
U705
IF_CON_SENSE
R777
R778
R779
C783
C789
C794


TR703
TR704
U701
U702
U705
U519
TR705
U518
U502
D501
D502
ZD501
BH103
IFC500

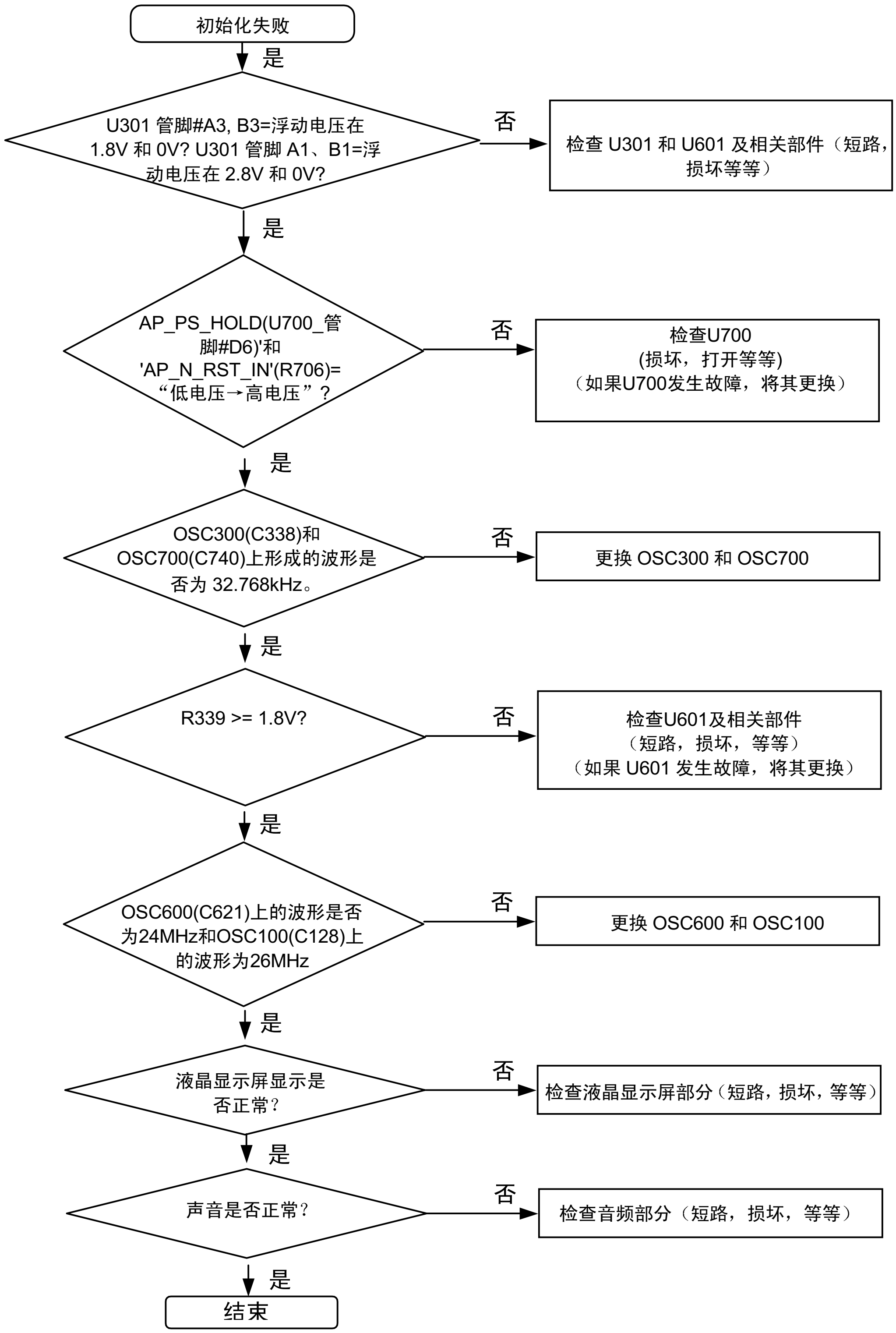

U700
OSC700
USB_VBUS_AP
USB_VBUS_CP
IF_TXD
IF_RXD
CP_TXD_2.8V
CP_RXD_2.8V
AP_TXD
AP_RXD
UART_SEL
32KHZ_AP
AP_N_RST_IN
nPOWER
AP_PMIC_IRQ
GPIO_nRST
AP_PMIC_SCL
AP_PMIC_SDA
VUSB_5V
IF_CON_SENSE
D600
VCC_2.8V_PDA

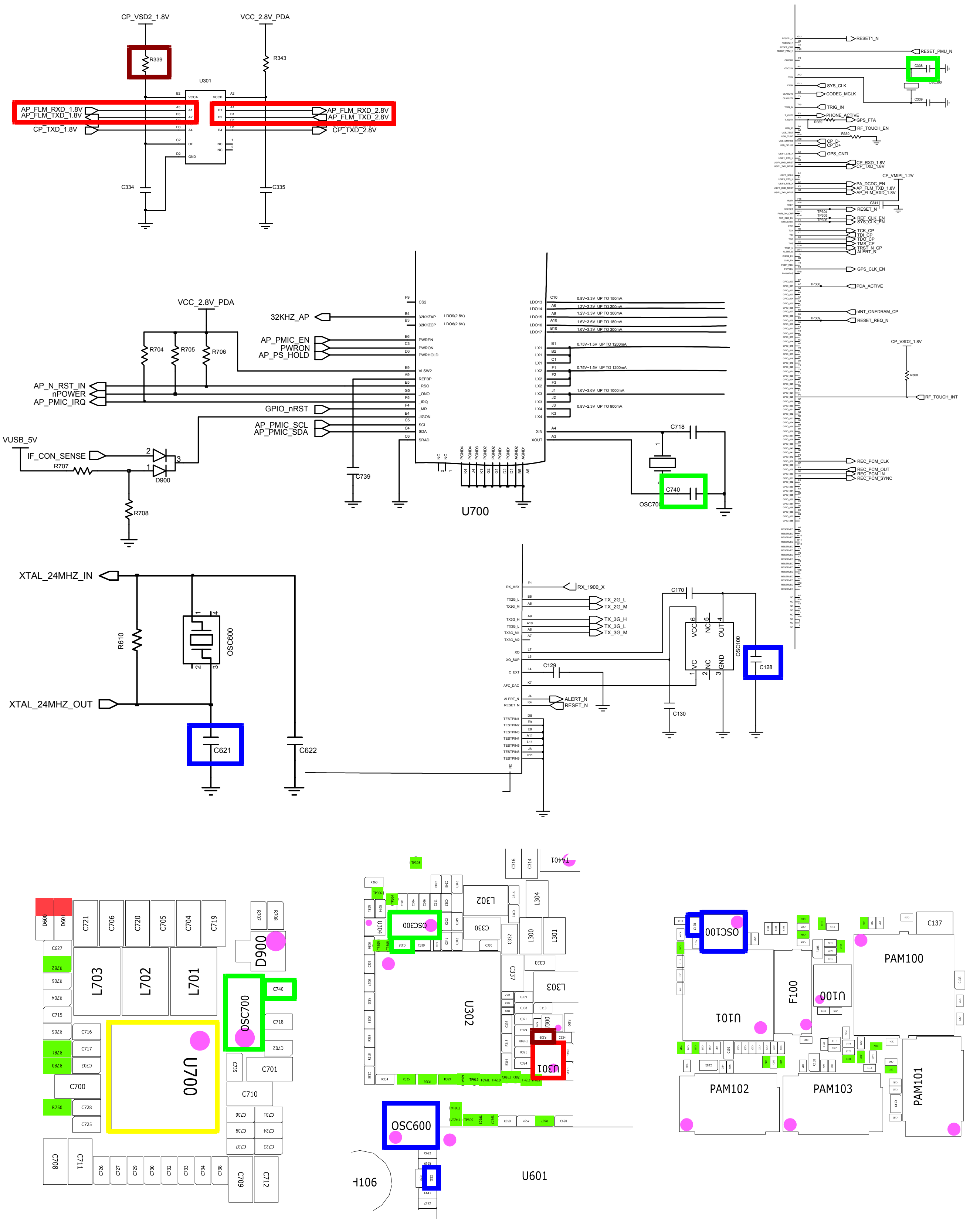

C718
C740
OSC700


R320
R326
ZD300
D301
D303
R322
R340
C340
R341
R325

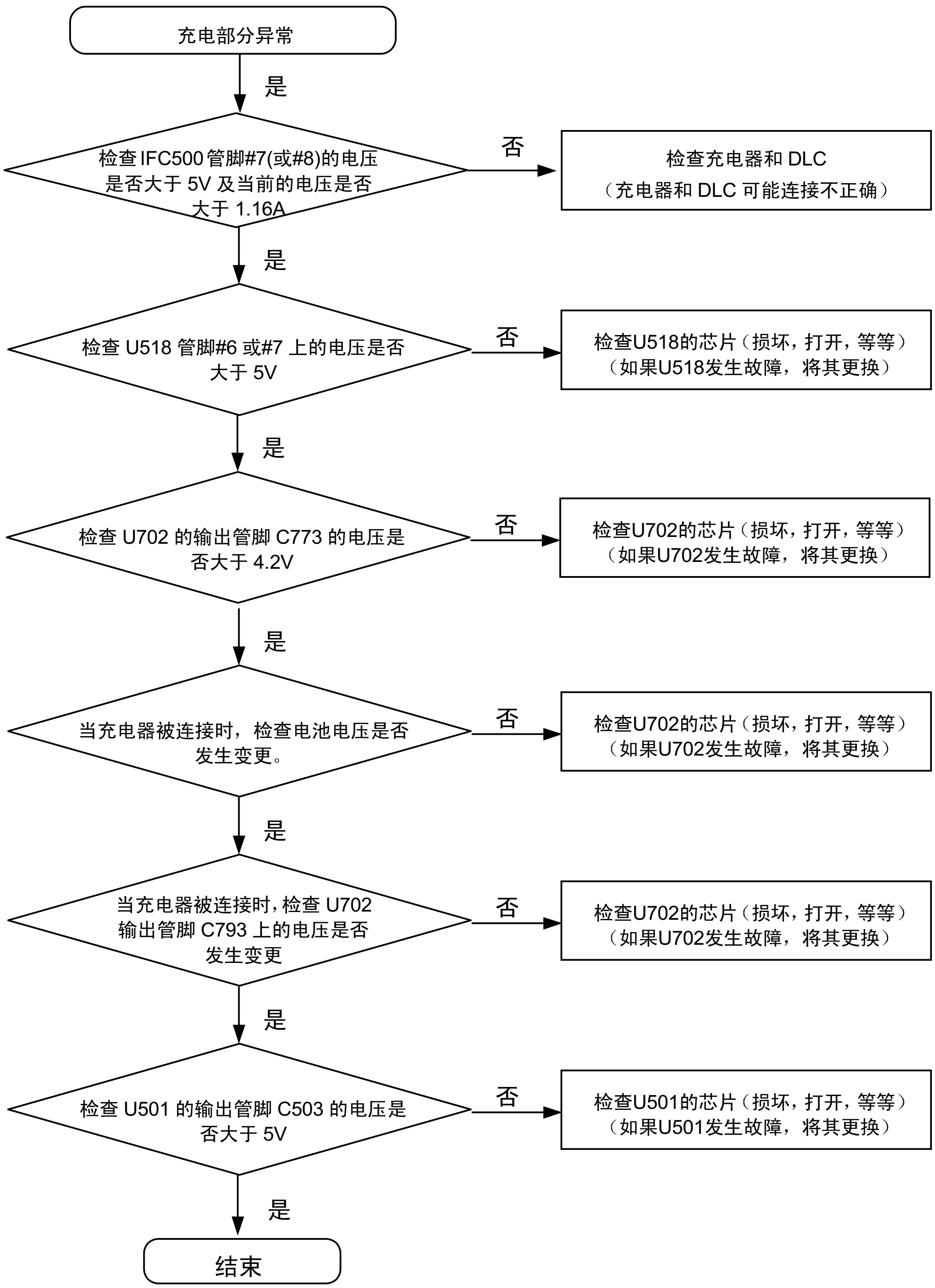

D600
C721
C706
C720
C705
C704
C719
R707
R708
D900
L703
L702
L701
OSC700
C740
C718
U700
C700
C701
C710
C708
C711
C709
C712

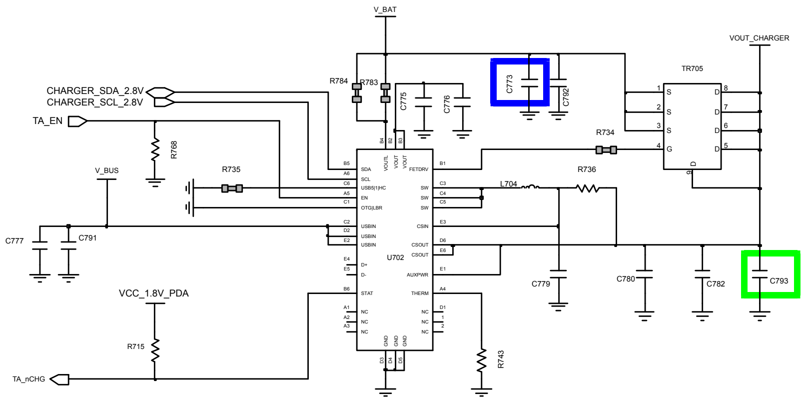
8-3-2. 初始化

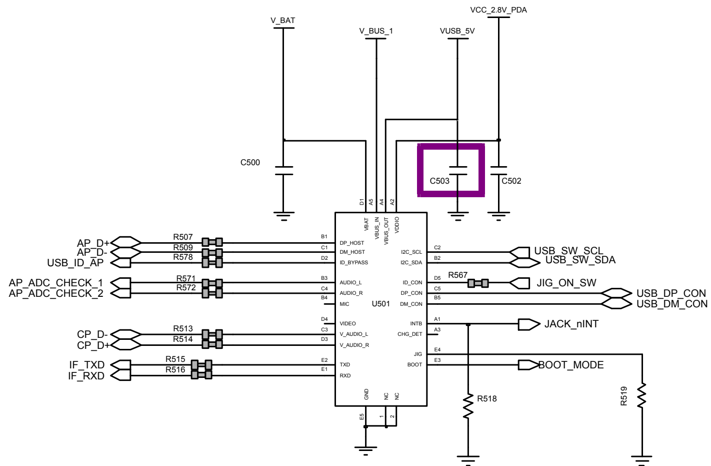

初始化失败
是
U301 管脚#A3, B3=浮动电压在 1.8V 和 0V? U301 管脚 A1、B1=浮动电压在 2.8V 和 0V?
否
检查 U301 和 U601 及相关部件（短路，损坏等等）
是
AP_PS_HOLD(U700_管脚#D6)'和 'AP_N_RST_IN'(R706)= “低电压→高电压”？
否
检查U700 (损坏，打开等等) （如果U700发生故障，将其更换）
是
OSC300(C338)和 OSC700(C740)上形成的波形是否为 32.768kHz。
否
更换 OSC300 和 OSC700
是
R339 >= 1.8V?
否
检查U601及相关部件 （短路，损坏，等等） （如果 U601 发生故障，将其更换）
是
OSC600(C621)上的波形是否为24MHz和OSC100(C128)上的波形为26MHz
否
更换 OSC600 和 OSC100
是
液晶显示屏显示是否正常？
否
检查液晶显示屏部分（短路，损坏，等等）
是
声音是否正常？
否
检查音频部分（短路，损坏，等等）
是
结束

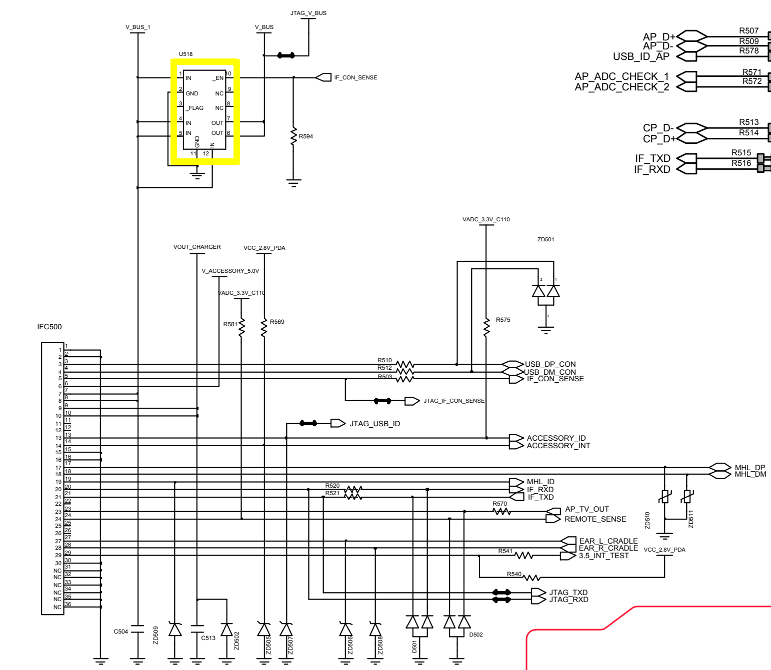

CP_VSD2_1.8V
VCC_2.8V_PDA
R339
R343
U301
AP_FLM_RXD_1.8V
AP_FLM_TXD_1.8V
CP_TXD_1.8V
AP_FLM_RXD_2.8V
AP_FLM_TXD_2.8V
CP_TXD_2.8V
C334
C335
32KHZ_AP
AP_PMIC_EN
PWRON
AP_PS_HOLD
R704
R705
R706
AP_N_RST_IN
nPOWER
AP_PMIC_IRQ
GPIO_nRST
AP_PMIC_SCL
AP_PMIC_SDA
VUSB_5V
IF_CON_SENSE
R707
D900
R708
C739
U700
C718
C740
OSC700
XTAL_24MHZ_IN
XTAL_24MHZ_OUT
R610
OSC600
C621
C622
RX_1900_X
TX_2G_L
TX_2G_M
TX_3G_H
TX_3G_L
TX_3G_M
C170
C129
C130
C128
OSC100
ALERT_N
RESET_N
RESET1_N
RESET_PMU_N
SYS_CLK
CODEC_MCLK
TRIG_IN
PHONE_ACTIVE
GPS_FTA
RF_TOUCH_EN
CP_D-
CP_D+
GPS_CNTL
CP_RXD_1.8V
CP_TXD_1.8V
CP_VMIPI_1.2V
PA_DCDC_EN
AP_FLM_TXD_1.8V
AP_FLM_RXD_1.8V
REF_CLK_EN
SYS_CLK_EN
TCK_CP
TDI_CP
TDO_CP
TMS_CP
TRST_N_CP
ALERT_N
GPS_CLK_EN
PDA_ACTIVE
nINT_ONEDRAM_CP
RESET_REQ_N
CP_VSD2_1.8V
R360
RF_TOUCH_INT
REC_PCM_CLK
REC_PCM_OUT
REC_PCM_IN
REC_PCM_SYNC
L703
L702
L701
D900
OSC700
U700
C700
C701
C710
OSC300
L302
L304
U302
L303
U301
OSC600
U601
OSC100
C137
PAM100
F100
U100
U101
PAM102
PAM103
PAM101

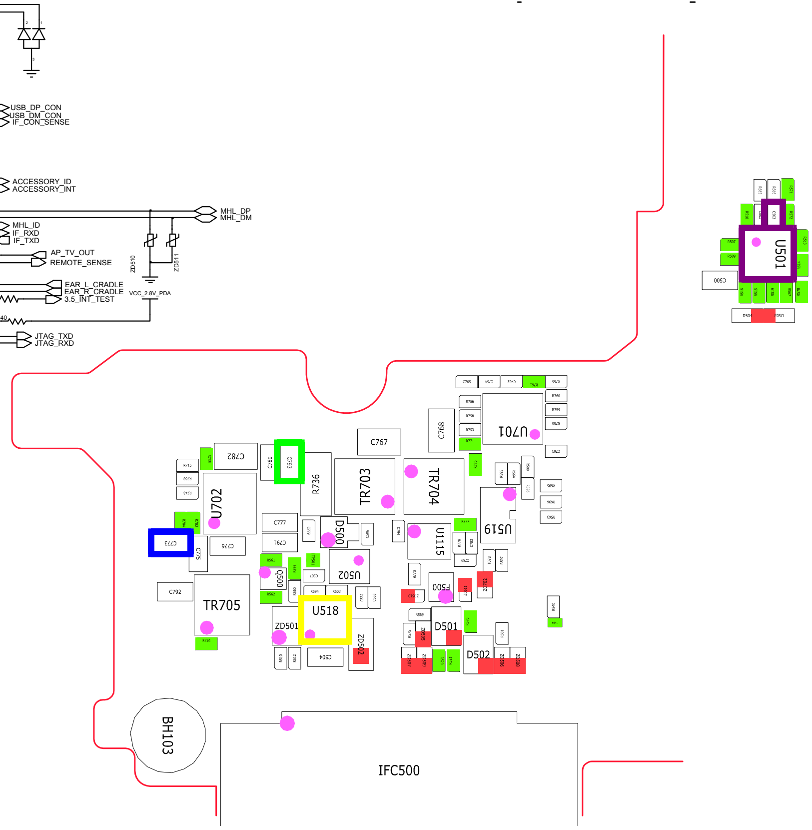
8-3-3.充电部分


充电部分异常
是
检查IFC500管脚#7(或#8)的电压是否大于 5V 及当前的电压是否大于 1.16A
否
检查充电器和 DLC
（充电器和 DLC 可能连接不正确）
是
检查 U518 管脚#6 或#7 上的电压是否大于 5V
否
检查U518的芯片（损坏，打开，等等）
（如果U518发生故障，将其更换）
是
检查 U702 的输出管脚 C773 的电压是否大于 4.2V
否
检查U702的芯片（损坏，打开，等等）
（如果U702发生故障，将其更换）
是
当充电器被连接时，检查电池电压是否发生变更。
否
检查U702的芯片（损坏，打开，等等）
（如果U702发生故障，将其更换）
是
当充电器被连接时，检查 U702 输出管脚 C793 上的电压是否发生变更
否
检查U702的芯片（损坏，打开，等等）
（如果U702发生故障，将其更换）
是
检查 U501 的输出管脚 C503 的电压是否大于 5V
否
检查U501的芯片（损坏，打开，等等）
（如果U501发生故障，将其更换）
是
结束

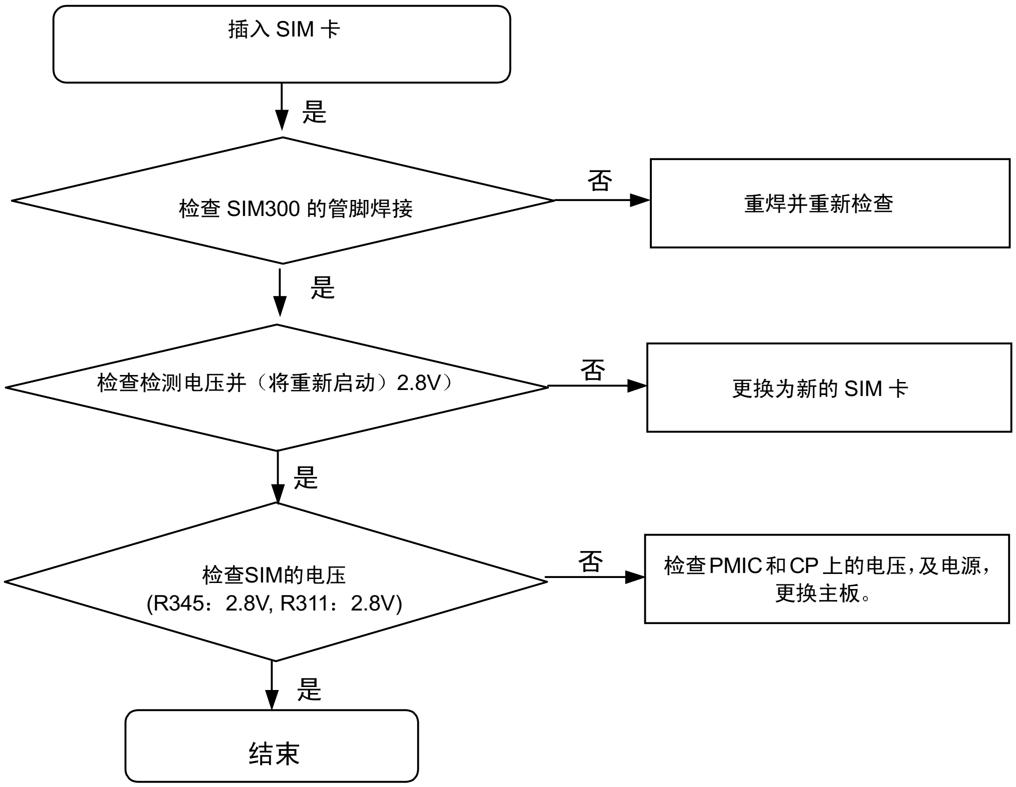

V_BAT
VOUT_CHARGER
TR705
CHARGER_SDA_2.8V
CHARGER_SCL_2.8V
TA_EN
R784
R783
C775
C776
C773
C792
R768
V_BUS
R735
R734
L704
R736
U702
C777
C791
VCC_1.8V_PDA
R715
TA_nCHG
C779
C780
C782
C793
R743

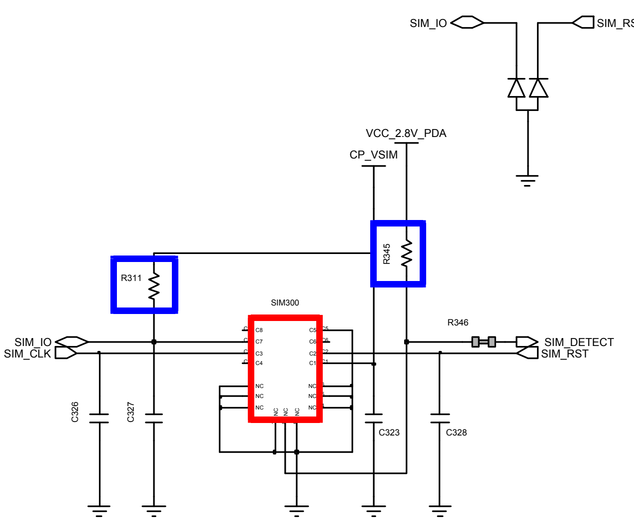

V_BAT
V_BUS_1
VUSB_5V
VCC_2.8V_PDA
C500
C503
C502
AP_D+
AP_D-
USB_ID_AP
R507
R509
R578
AP_ADC_CHECK_1
AP_ADC_CHECK_2
R571
R572
U501
CP_D-
CP_D+
R513
R514
IF_TXD
IF_RXD
R515
R516
USB_SW_SCL
USB_SW_SDA
R567
JIG_ON_SW
USB_DP_CON
USB_DM_CON
JACK_nINT
BOOT_MODE
R518
R519

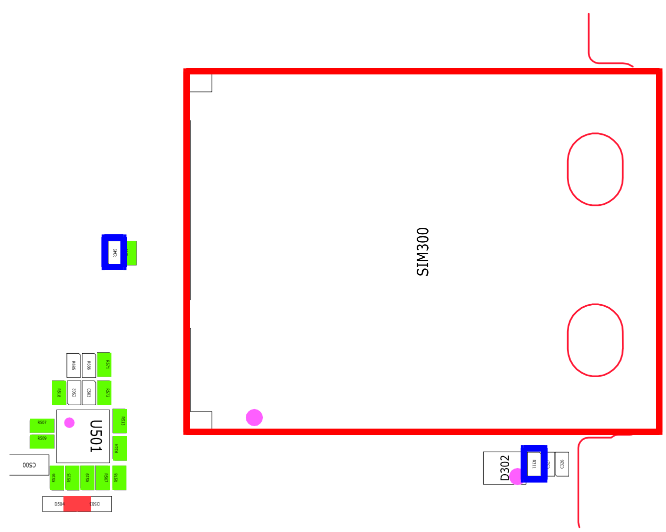

V_BUS_1
V_BUS
JTAG_V_BUS
U518
IF_CON_SENSE
R594
VADC_3.3V_C110
ZD501
VOUT_CHARGER
VCC_2.8V_PDA
V_ACCESSORY_5.0V
IFC500
R581
R569
R575
R510
R512
USB_DP_CON
USB_DM_CON
IF_CON_SENSE
JTAG_IF_CON_SENSE
JTAG_USB_ID
ACCESSORY_ID
ACCESSORY_INT
MHL_DP
MHL_DM
R520
R521
MHL_ID
IF_RXD
IF_TXD
R570
AP_TV_OUT
REMOTE_SENSE
EAR_L_CRADLE
EAR_R_CRADLE
3.5_INT_TEST
R541
R540
VCC_2.8V_PDA
JTAG_TXD
JTAG_RXD
C504
C513
D502


U501
C500
U701
C767
C768
TR703
TR704
U702
C782
C793
R736
C773
C777
D500
U1115
U519
C792
TR705
U502
F500
U518
ZD501
D501
D502
C504
BH103
IFC500

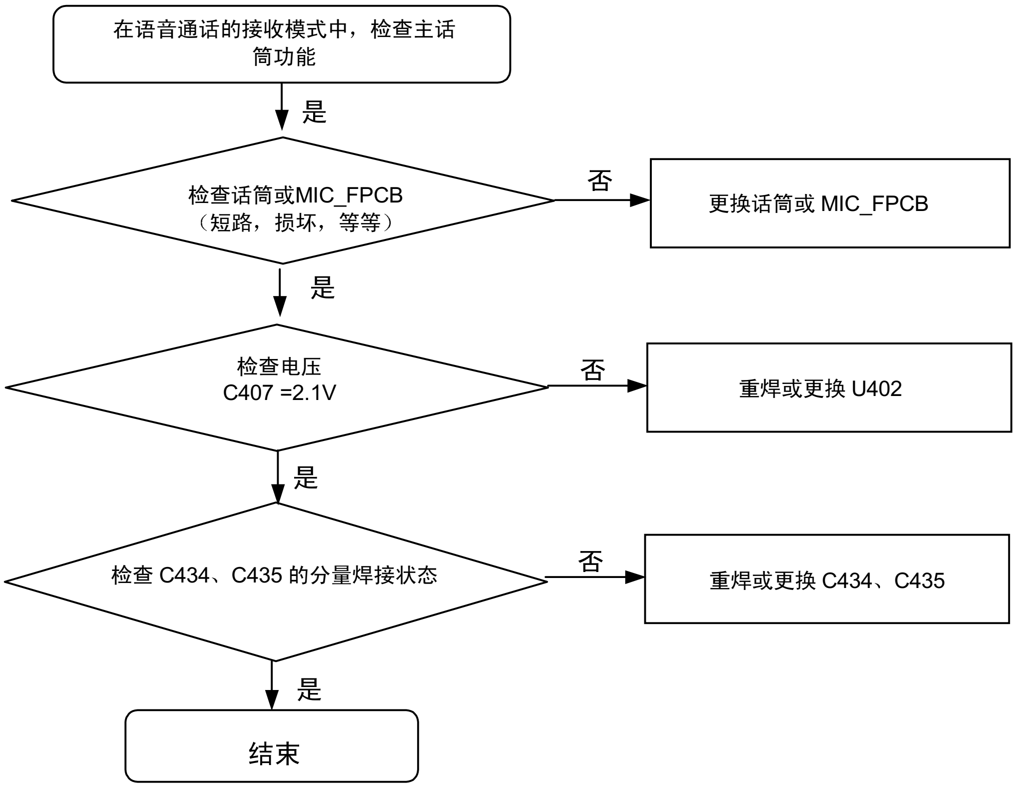
8-3-4.Sim部分

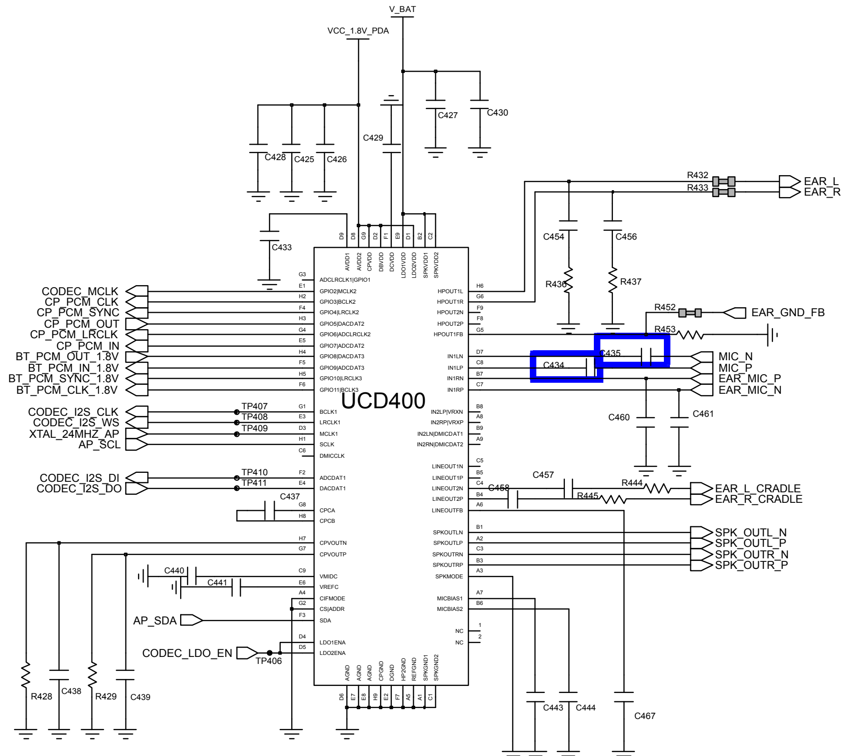

插入 SIM 卡
是
检查 SIM300 的管脚焊接
否
重焊并重新检查
是
检查检测电压并（将重新启动）2.8V）
否
更换为新的 SIM 卡
是
检查SIM的电压
(R345：2.8V, R311：2.8V)
否
检查PMIC和CP上的电压，及电源，
更换主板。
是
结束

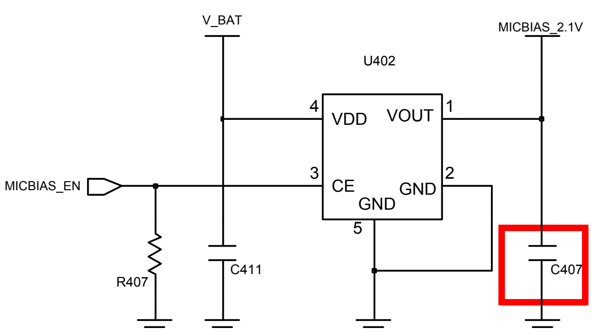

SIM_IO
SIM_RST
VCC_2.8V_PDA
CP_VSIM
R311
R345
SIM300
R346
SIM_IO
SIM_CLK
SIM_DETECT
SIM_RST
C326
C327
C323
C328

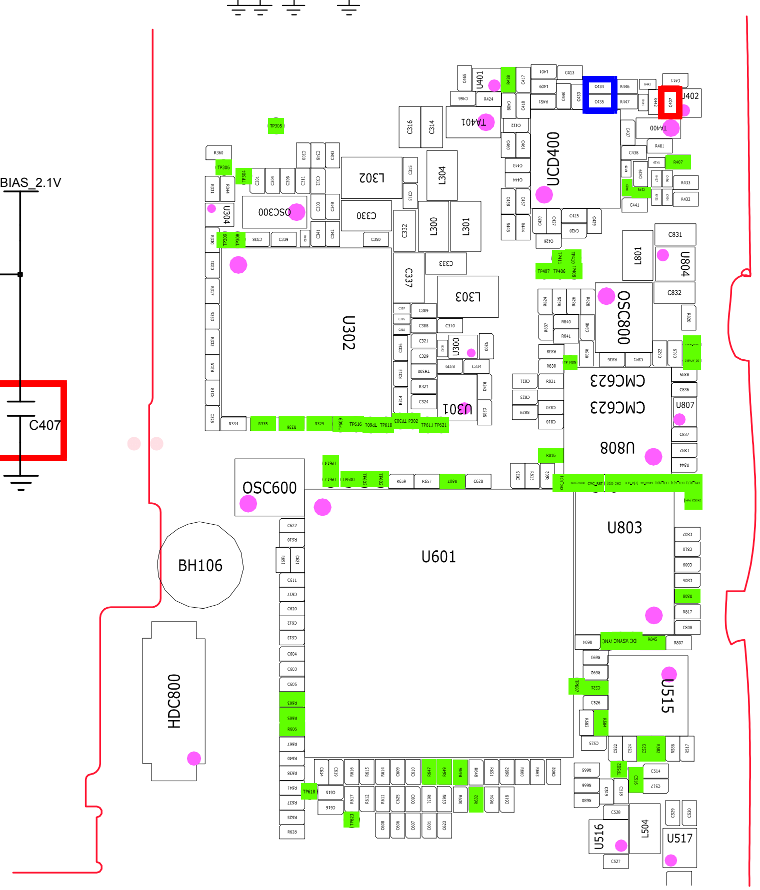

SIM300
U501
D302

8-3-5.话筒部分

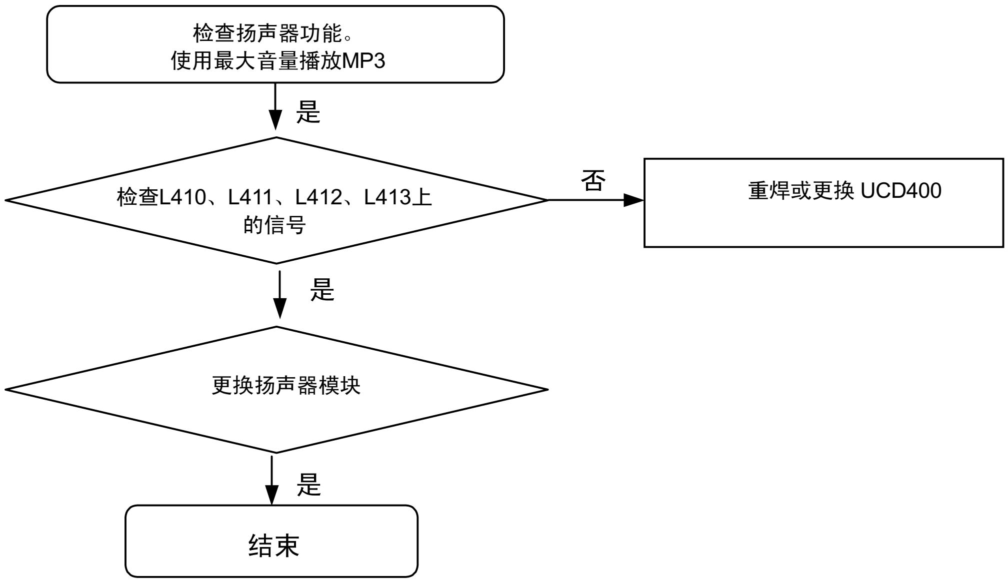

在语音通话的接收模式中，检查主话筒功能
是
检查话筒或MIC_FPCB
（短路，损坏，等等）
否
更换话筒或 MIC_FPCB
是
检查电压
C407 =2.1V
否
重焊或更换 U402
是
检查 C434、C435 的分量焊接状态
否
重焊或更换 C434、C435
是
结束

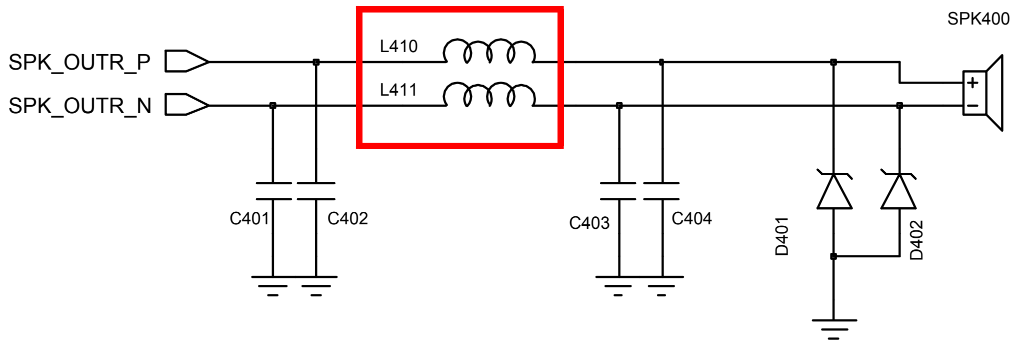

V_BAT
VCC_1.8V_PDA
UCD400
C427
C430
C429
C428
C425
C426
C433
R432
R433
EAR_L
EAR_R
C454
C456
R436
R437
R452
R453
EAR_GND_FB
C435
C434
MIC_N
MIC_P
EAR_MIC_P
EAR_MIC_N
C460
C461
CODEC_MCLK
CP_PCM_CLK
CP_PCM_SYNC
CP_PCM_OUT
CP_PCM_LRCLK
CP_PCM_IN
BT_PCM_OUT_1.8V
BT_PCM_IN_1.8V
BT_PCM_SYNC_1.8V
BT_PCM_CLK_1.8V
CODEC_I2S_CLK
CODEC_I2S_WS
XTAL_24MHZ_AP
AP_SCL
TP407
TP408
TP409
CODEC_I2S_DI
CODEC_I2S_DO
TP410
TP411
C437
C457
C458
R444
R445
EAR_L_CRADLE
EAR_R_CRADLE
SPK_OUTL_N
SPK_OUTL_P
SPK_OUTR_N
SPK_OUTR_P
C440
C441
AP_SDA
CODEC_LDO_EN
TP406
R428
C438
R429
C439
C443
C444
C467

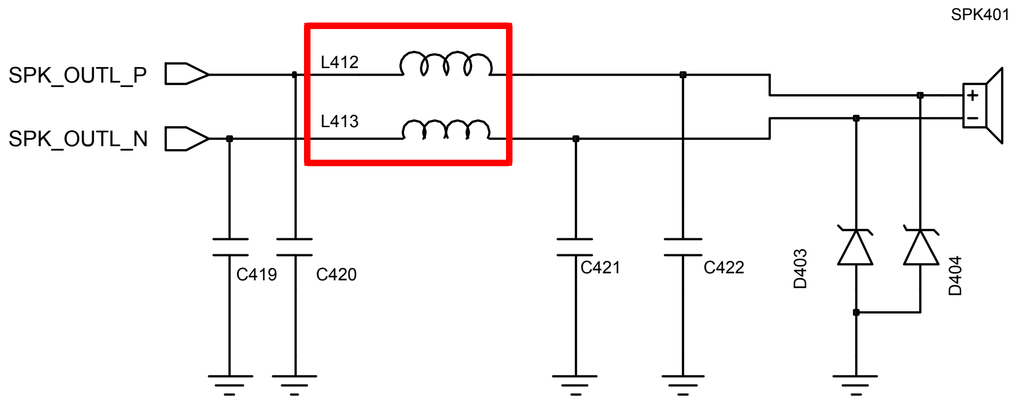

V_BAT
MICBIAS_2.1V
U402
4
VDD
VOUT
1
3
CE
GND
2
GND
5
MICBIAS_EN
R407
C411
C407

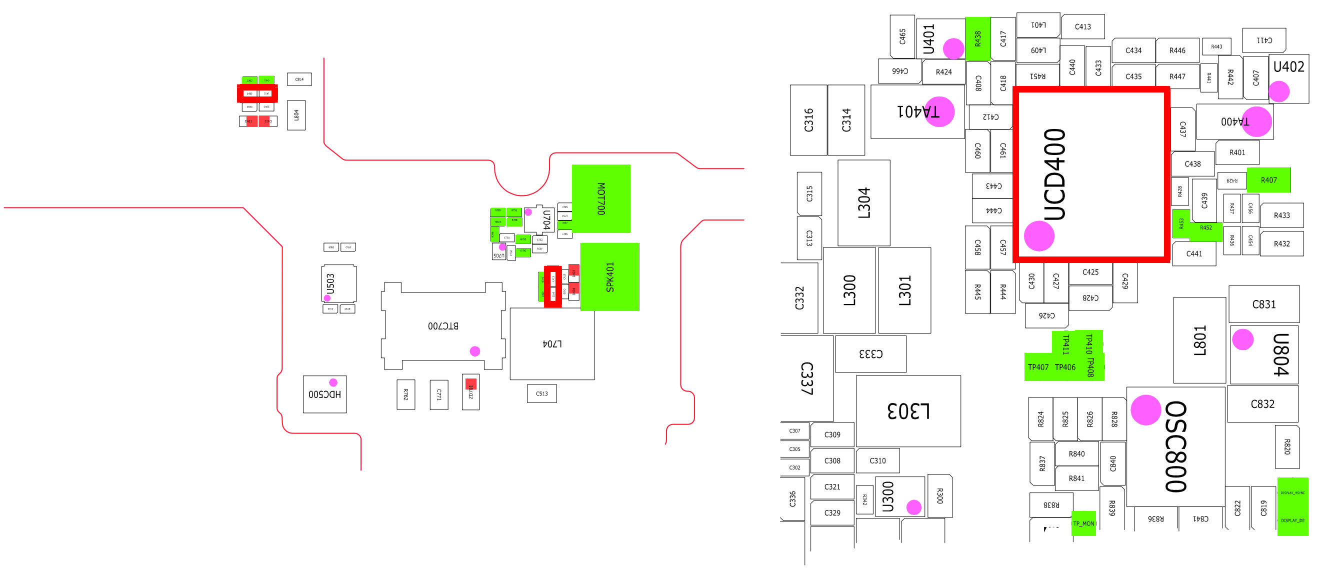

UCD400
U401
U402
U302
U304
OSC300
C330
L302
L304
L300
L301
L303
U300
U301
OSC800
CMC623
CMC623
U808
U804
U807
OSC600
U601
U803
BH106
HDC800
U515
U516
L504
U517

8-3-6.扬声器部分

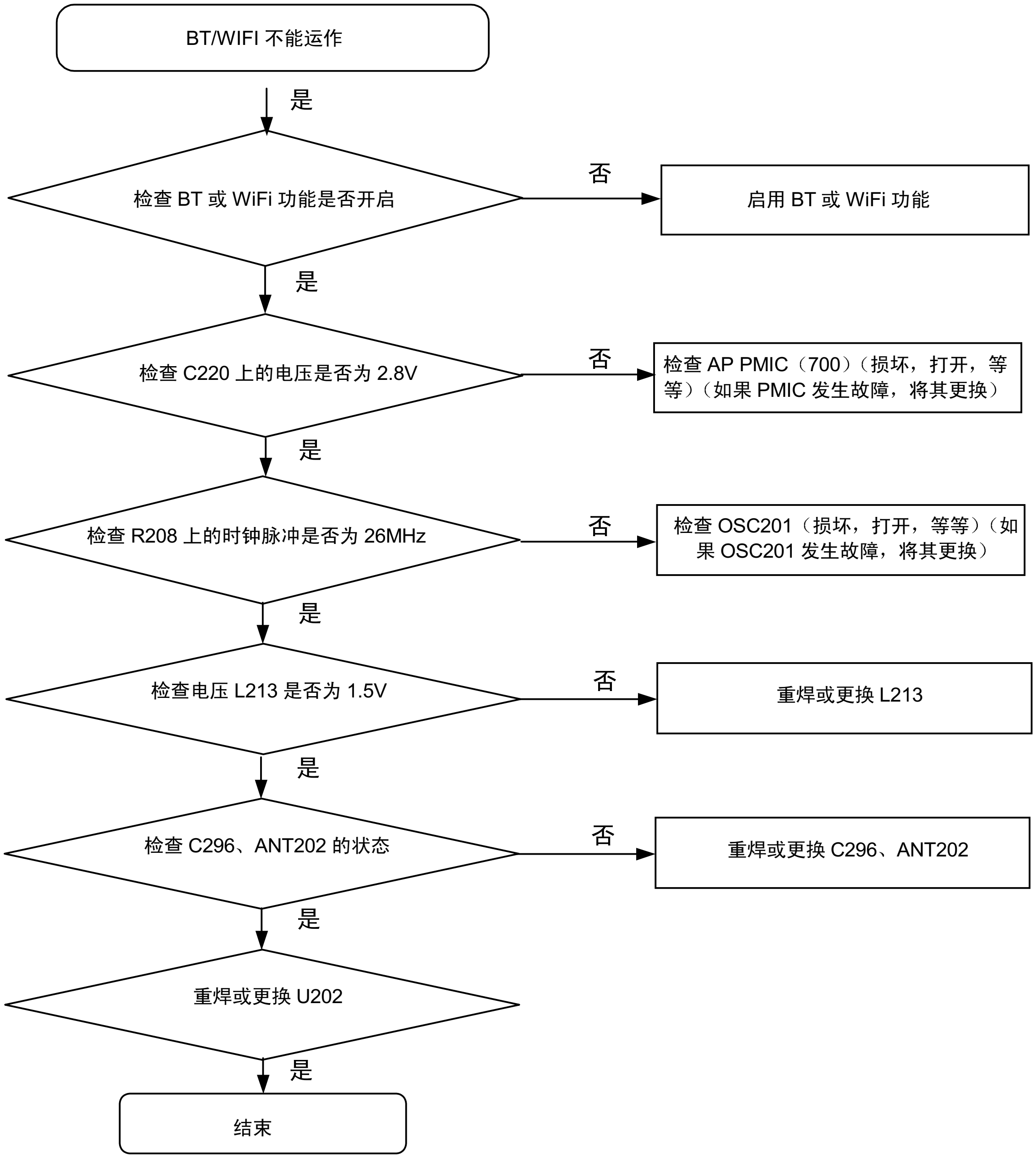

检查扬声器功能。
使用最大音量播放MP3
是
检查L410、L411、L412、L413上
的信号
否
重焊或更换 UCD400
是
更换扬声器模块
是
结束

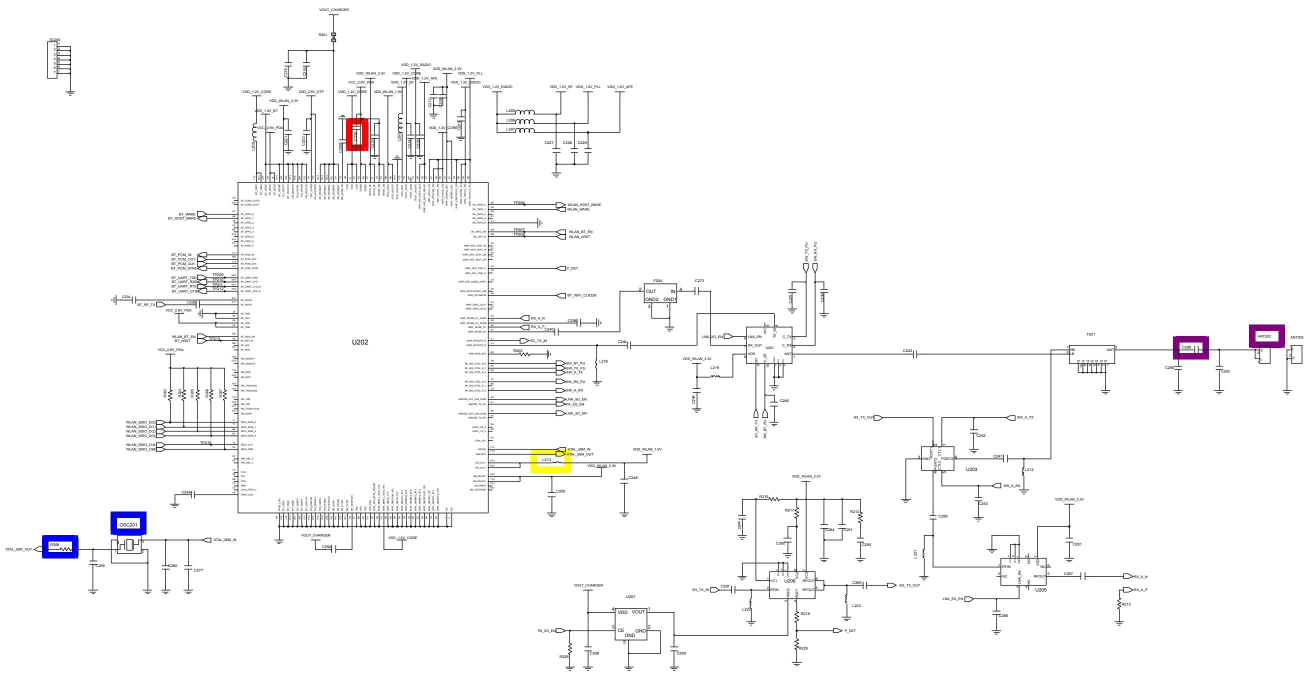

SPK400
SPK_OUTR_P
SPK_OUTR_N
L410
L411
C401
C402
C403
C404
D401
D402

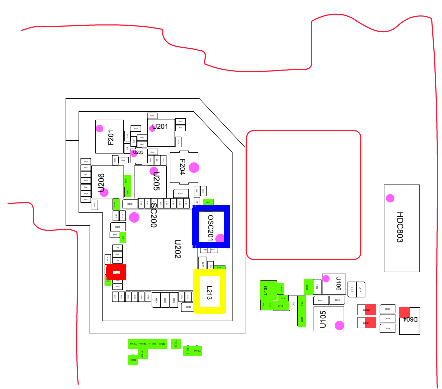

SPK401
SPK_OUTL_P
SPK_OUTL_N
L412
L413
C419
C420
C421
C422
D403
D404

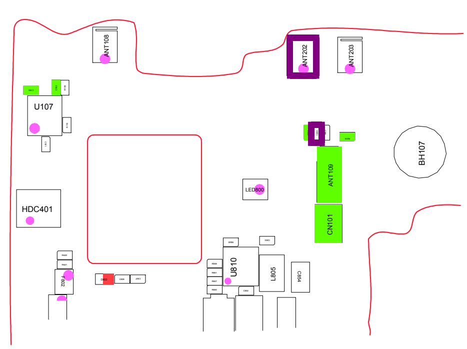

UCD400
U401
U402
TA401
TA400
U804
L801
OSC800
L303
C337
C333
L300
L301
L304
C316
C314
C332
U300
MOT700
SPK401
BTC700
L704
U503
HDC500
C831
C832

8-3-7 BT/WIFI

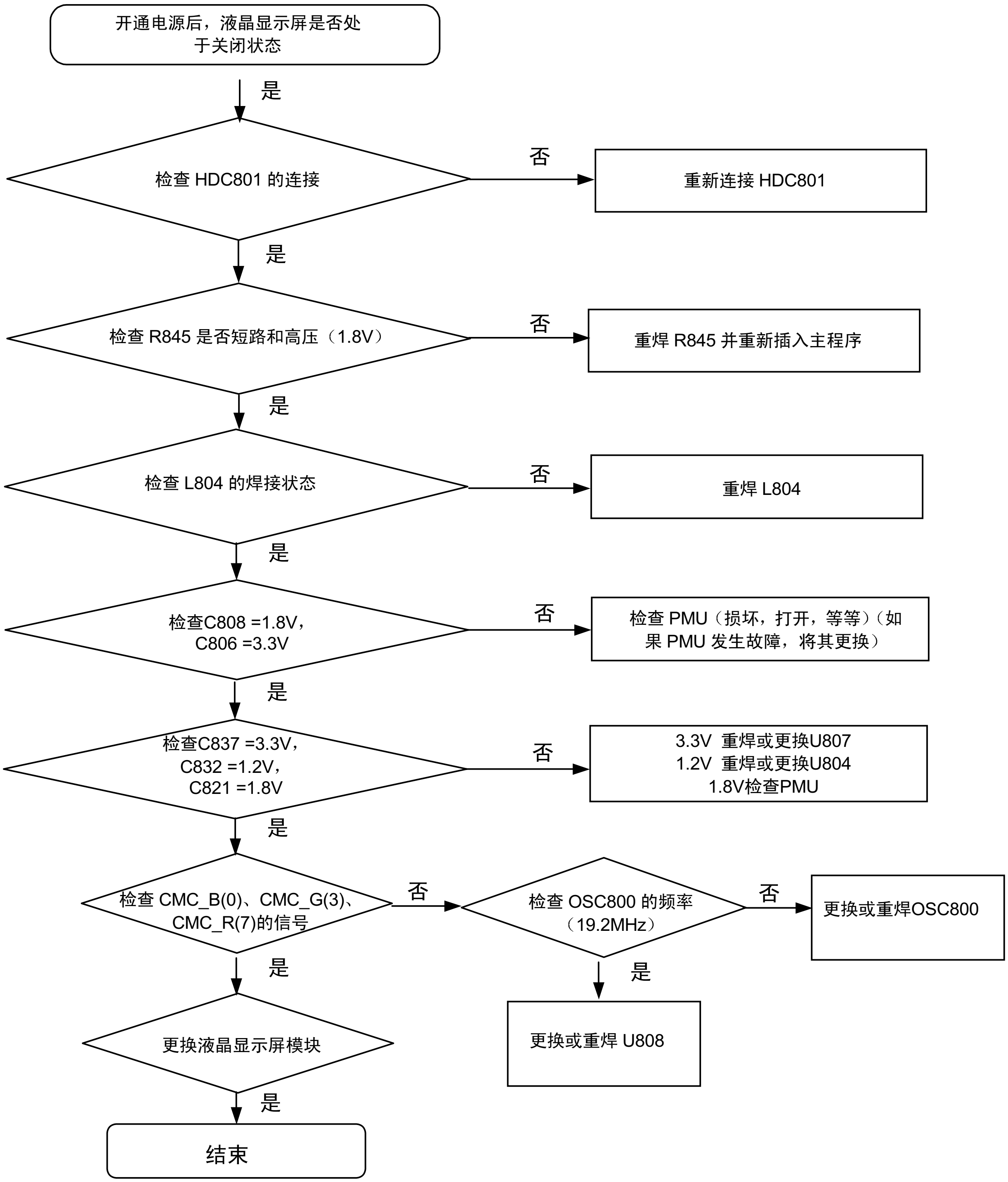

BT/WIFI 不能运作
是
检查 BT 或 WiFi 功能是否开启
否
启用 BT 或 WiFi 功能
是
检查 C220 上的电压是否为 2.8V
否
检查 AP PMIC（700）（损坏，打开，等等）（如果 PMIC 发生故障，将其更换）
是
检查 R208 上的时钟脉冲是否为 26MHz
否
检查 OSC201（损坏，打开，等等）（如果 OSC201 发生故障，将其更换）
是
检查电压 L213 是否为 1.5V
否
重焊或更换 L213
是
检查 C296、ANT202 的状态
否
重焊或更换 C296、ANT202
是
重焊或更换 U202
是
结束

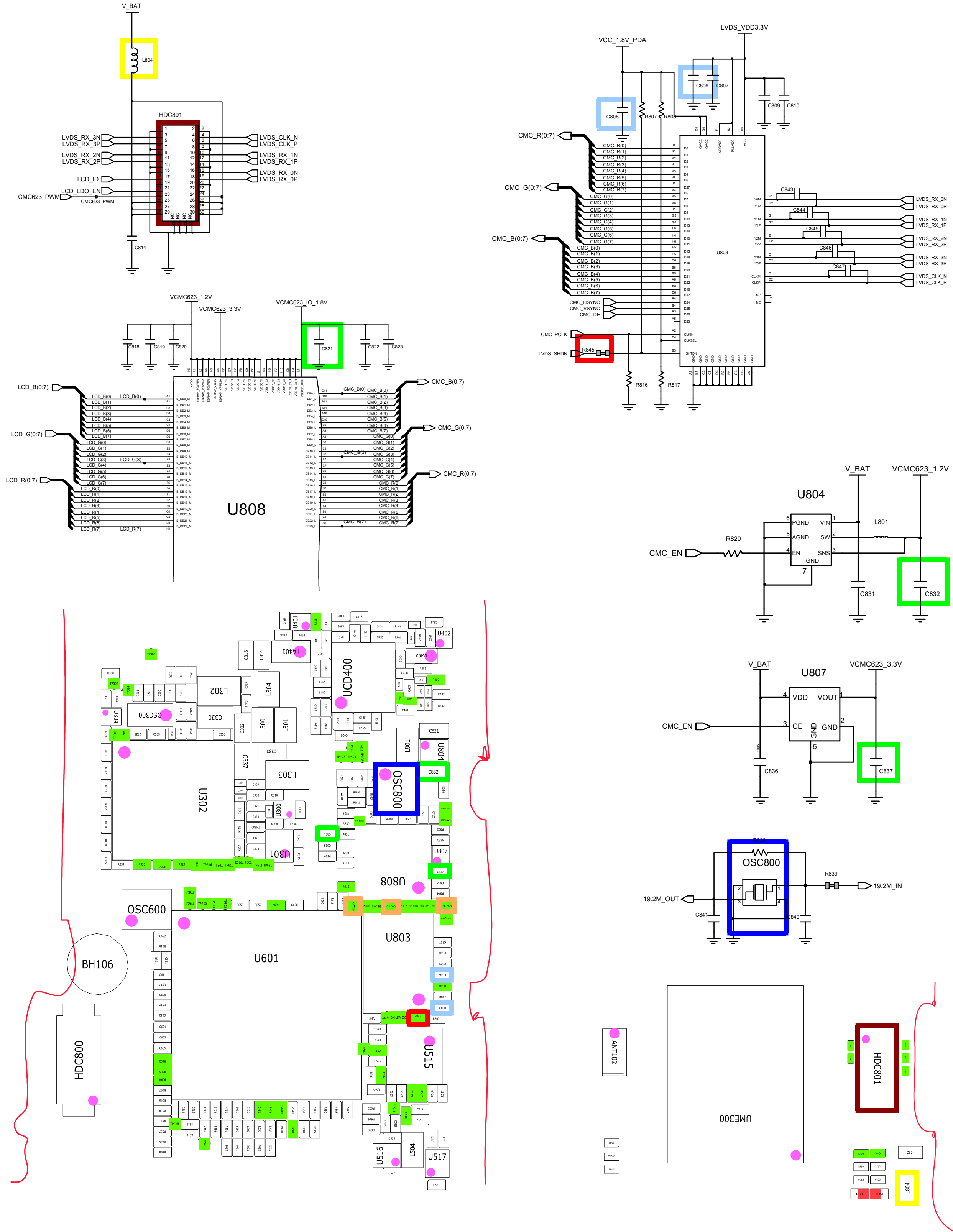

U202


F201
U201
U206
U205
F204
C200
OSC201
U202
L213
HDC803
U106
U105

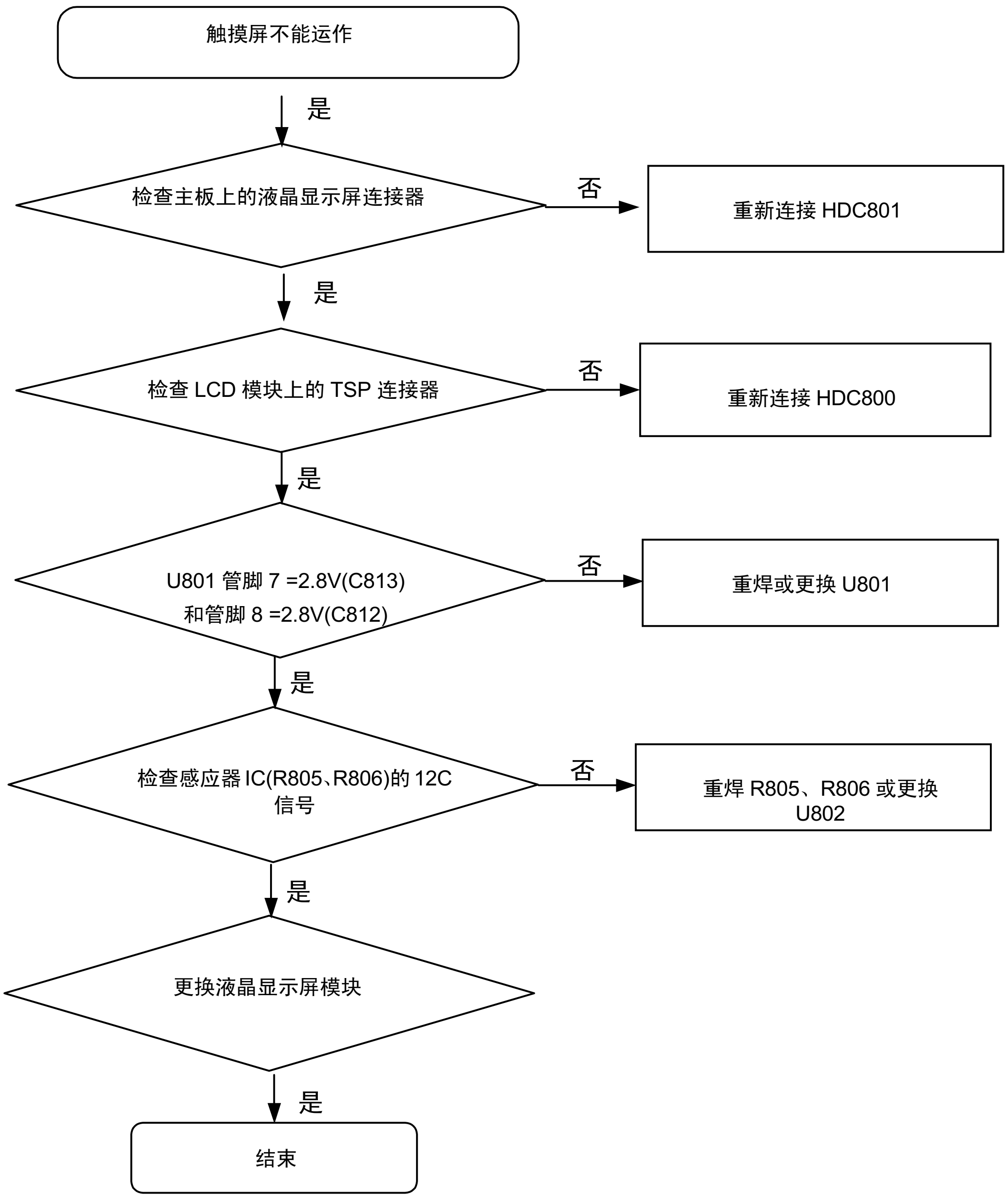

ANT108
ANT202
ANT203
U107
BH107
HDC401
ANT109
CN101
U810

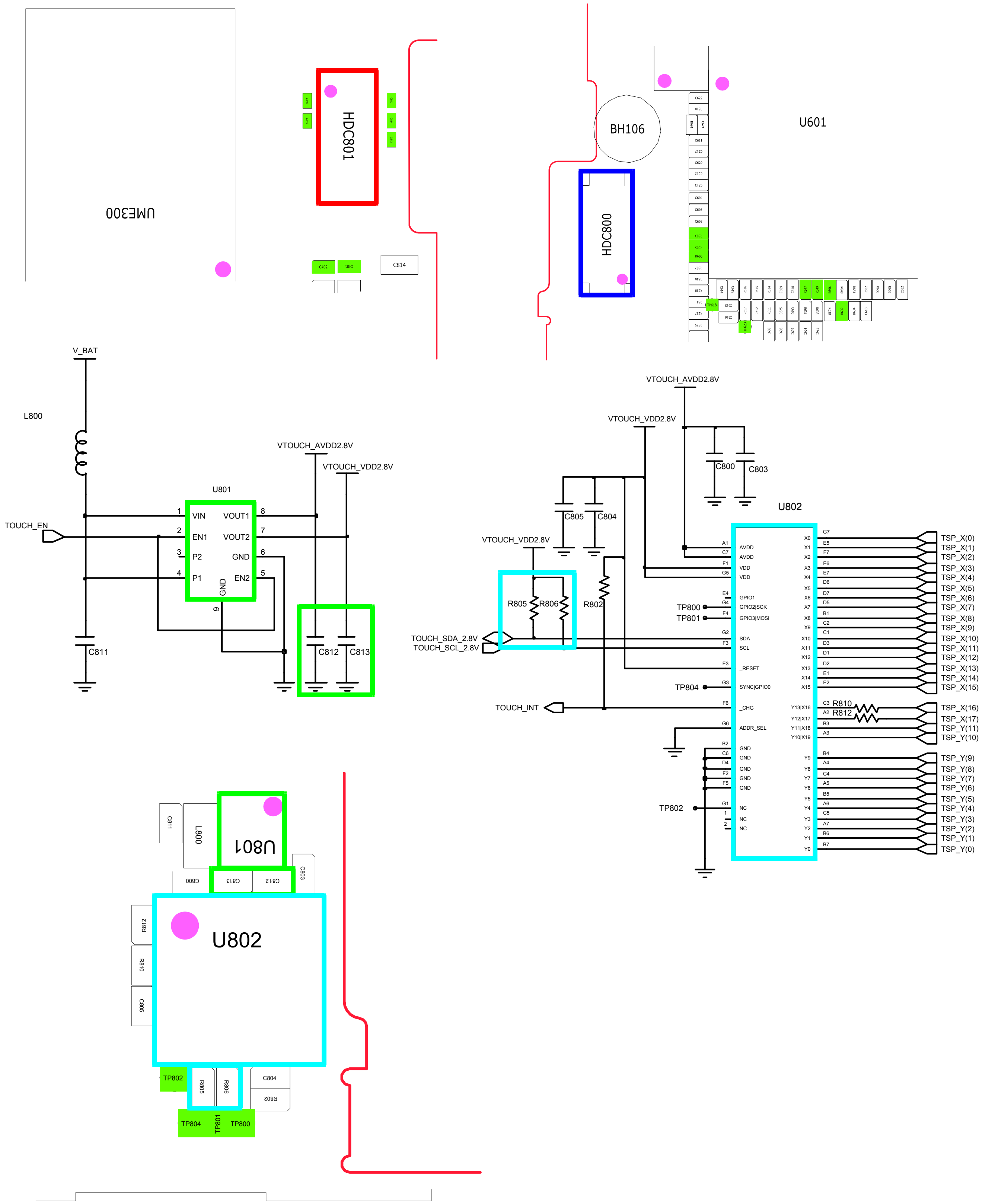
8-3-8.液晶显示屏


开通电源后，液晶显示屏是否处于关闭状态
是
检查 HDC801 的连接
否
重新连接 HDC801
是
检查 R845 是否短路和高压（1.8V）
否
重焊 R845 并重新插入主程序
是
检查 L804 的焊接状态
否
重焊 L804
是
检查C808 =1.8V，C806 =3.3V
否
检查 PMU（损坏，打开，等等）（如果 PMU 发生故障，将其更换）
是
检查C837 =3.3V，C832 =1.2V，C821 =1.8V
否
3.3V 重焊或更换U807
1.2V 重焊或更换U804
1.8V检查PMU
是
检查 CMC_B(0)、CMC_G(3)、CMC_R(7)的信号
否
检查 OSC800 的频率（19.2MHz）
否
更换或重焊OSC800
是
更换或重焊 U808
是
更换液晶显示屏模块
是
结束

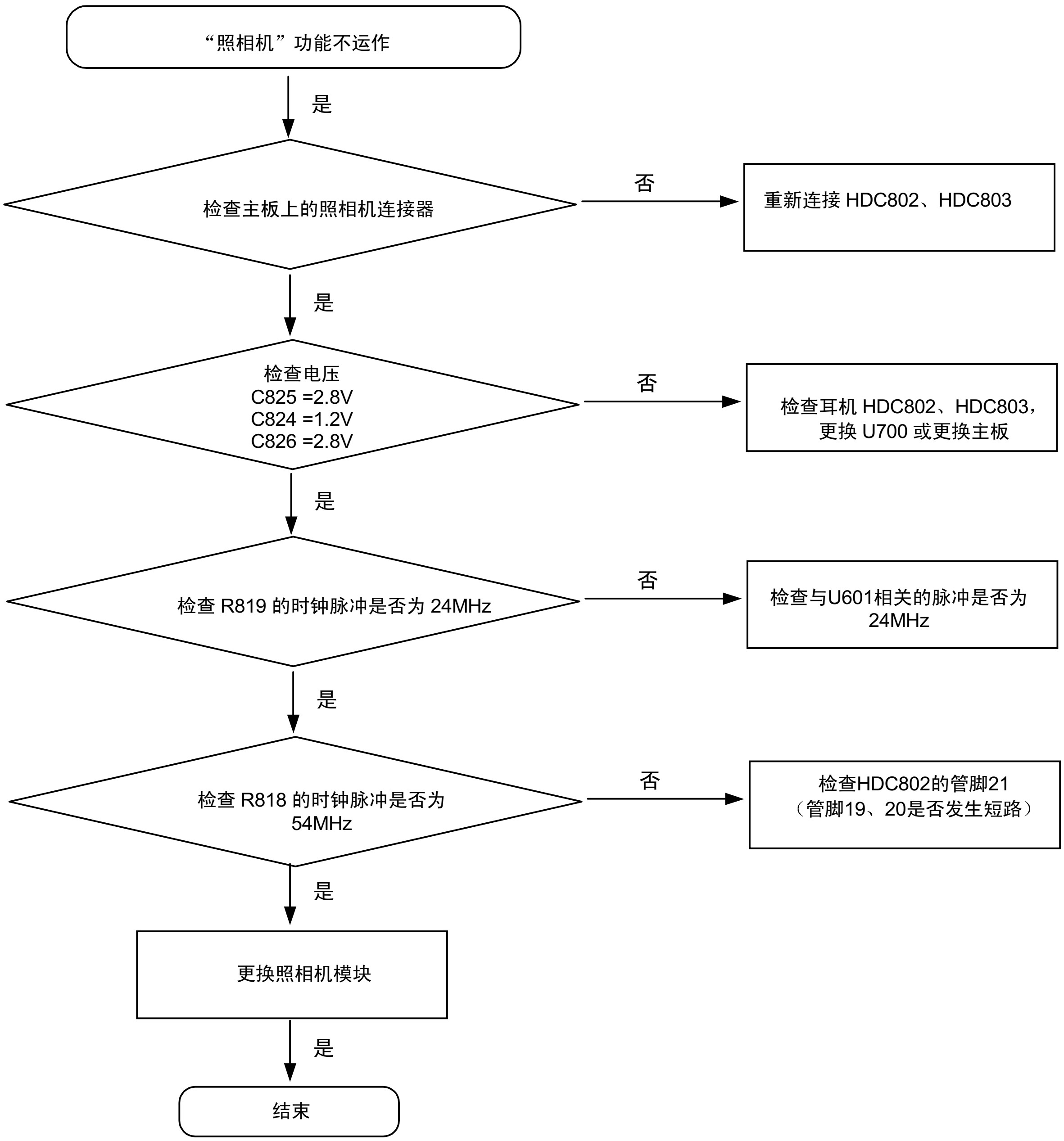

V_BAT
L804
HDC801
LVDS_RX_3N
LVDS_RX_3P
LVDS_RX_2N
LVDS_RX_2P
LCD_ID
LCD_LDO_EN
CMC623_PWM
LVDS_CLK_N
LVDS_CLK_P
LVDS_RX_1N
LVDS_RX_1P
LVDS_RX_0N
LVDS_RX_0P
C814
VCMC623_1.2V
VCMC623_3.3V
VCMC623_IO_1.8V
C818
C819
C820
C821
C822
C823
LCD_B(0:7)
LCD_G(0:7)
LCD_R(0:7)
CMC_B(0:7)
CMC_G(0:7)
CMC_R(0:7)
U808
VCC_1.8V_PDA
LVDS_VDD3.3V
C806
C807
C808
C809
C810
R807
R808
CMC_HSYNC
CMC_VSYNC
CMC_DE
CMC_PCLK
LVDS_SHDN
R845
R816
R817
U803
C843
C844
C845
C846
C847
V_BAT
VCMC623_1.2V
U804
PGND
VIN
AGND
SW
EN
SNS
GND
R820
CMC_EN
L801
C831
C832
V_BAT
VCMC623_3.3V
U807
VDD
VOUT
CE
GND
CMC_EN
C836
C837
OSC800
R839
19.2M_OUT
19.2M_IN
C841
C840
U302
OSC300
L302
C330
L304
L300
L301
C337
L303
U300
U301
UCD400
TA401
U402
L801
U804
OSC800
C832
U807
U808
U803
OSC600
U601
BH106
HDC800
U515
U516
L504
U517
ANT102
UME300
HDC801
L804
C814

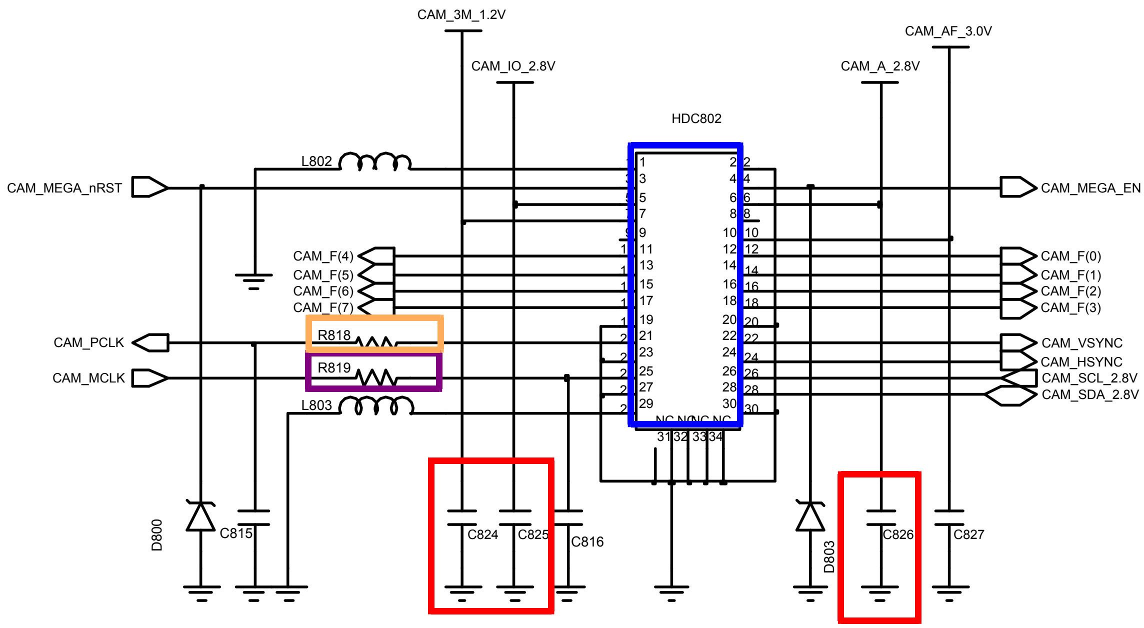
8-3-9.TSP

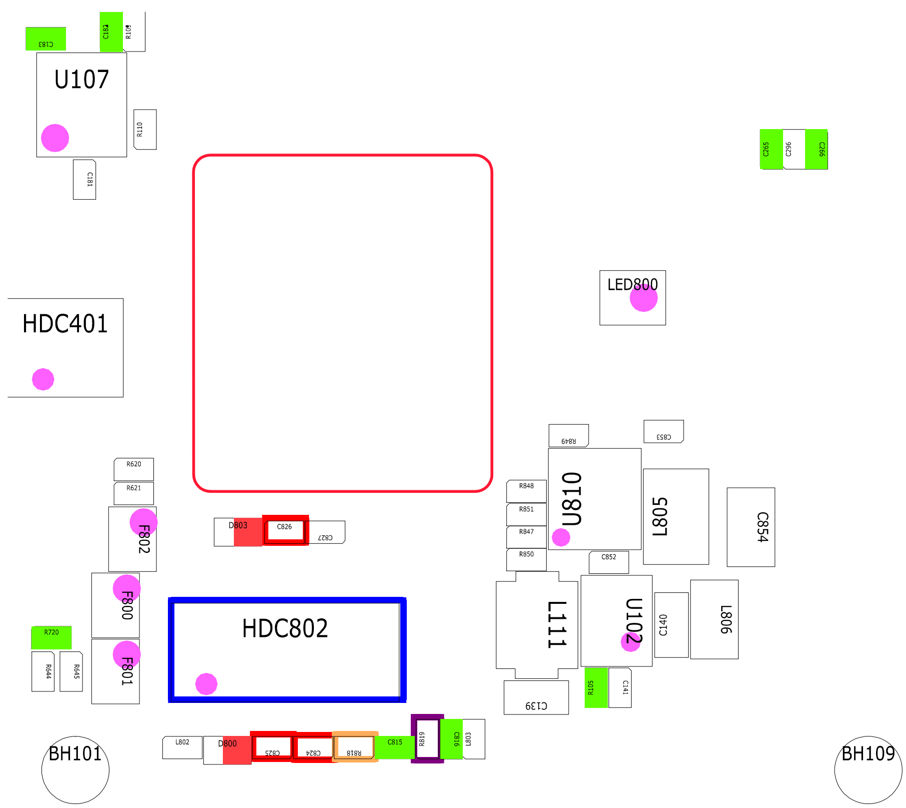

触摸屏不能运作
是
检查主板上的液晶显示屏连接器
否
重新连接 HDC801
是
检查 LCD 模块上的 TSP 连接器
否
重新连接 HDC800
是
U801 管脚 7 =2.8V(C813)
和管脚 8 =2.8V(C812)
否
重焊或更换 U801
是
检查感应器 IC(R805、R806)的 12C
信号
否
重焊 R805、R806 或更换
U802
是
更换液晶显示屏模块
是
结束


UME300
HDC801
BH106
U601
HDC800
C814
V_BAT
L800
TOUCH_EN
U801
VIN
EN1
P2
P1
VOUT1
VOUT2
GND
EN2
VTOUCH_AVDD2.8V
VTOUCH_VDD2.8V
C811
C812
C813
C800
C803
C805
C804
U802
R805
R806
R802
TP800
TP801
TP804
TP802
TOUCH_SDA_2.8V
TOUCH_SCL_2.8V
TOUCH_INT
AVDD
VDD
GPIO1
GPIO2|SCK
GPIO3|MOSI
SDA
SCL
_RESET
SYNC|GPIO0
_CHG
ADDR_SEL
NC
R810
R812
TSP_X(0)
TSP_X(1)
TSP_X(2)
TSP_X(3)
TSP_X(4)
TSP_X(5)
TSP_X(6)
TSP_X(7)
TSP_X(8)
TSP_X(9)
TSP_X(10)
TSP_X(11)
TSP_X(12)
TSP_X(13)
TSP_X(14)
TSP_X(15)
TSP_X(16)
TSP_X(17)
TSP_Y(11)
TSP_Y(10)
TSP_Y(9)
TSP_Y(8)
TSP_Y(7)
TSP_Y(6)
TSP_Y(5)
TSP_Y(4)
TSP_Y(3)
TSP_Y(2)
TSP_Y(1)
TSP_Y(0)

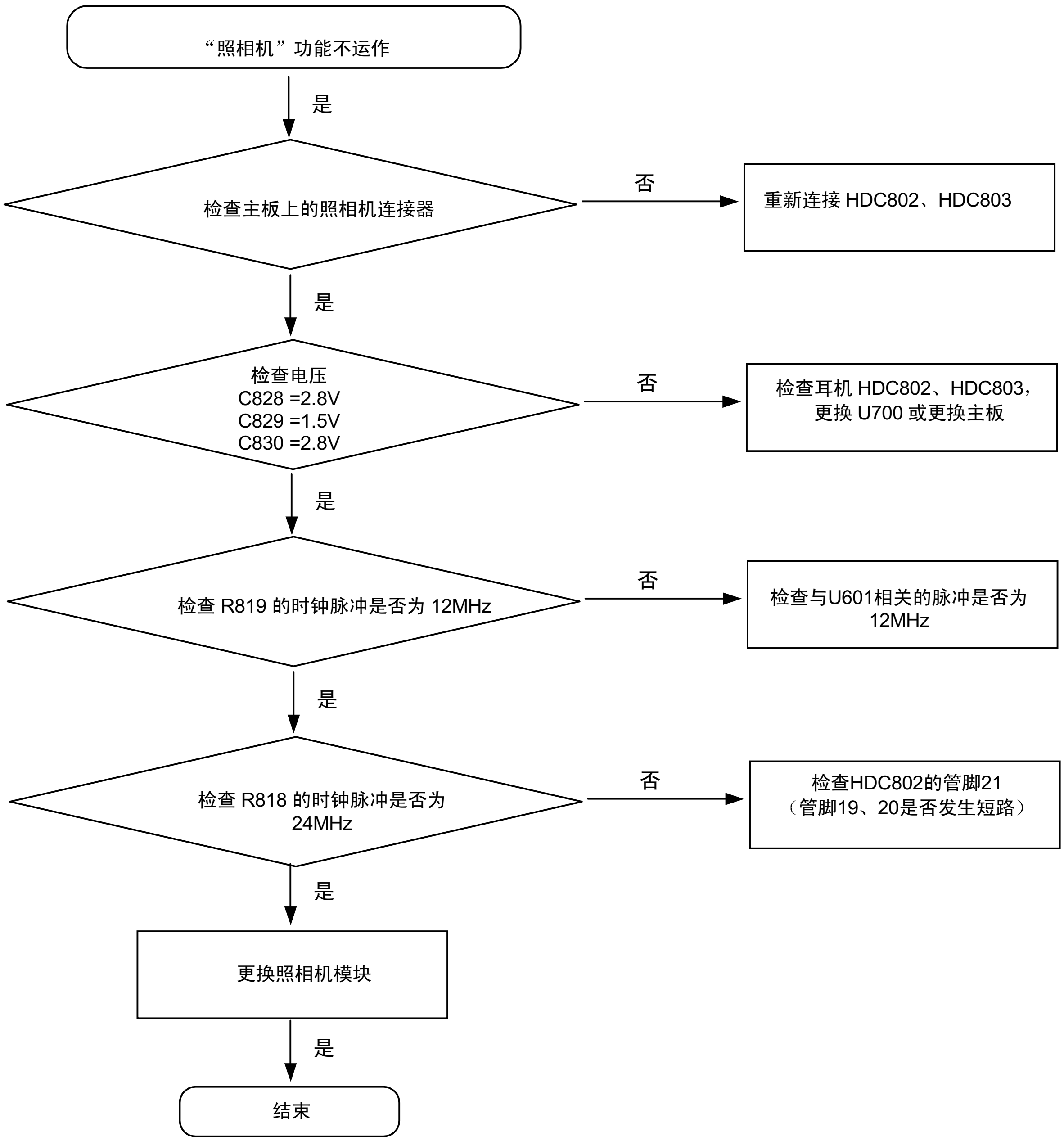
8-3-10.3M照相机

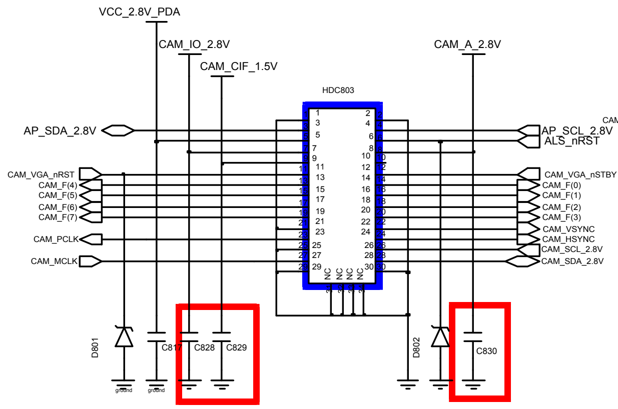

“照相机”功能不运作
是
检查主板上的照相机连接器
否
重新连接 HDC802、HDC803
是
检查电压
C825 =2.8V
C824 =1.2V
C826 =2.8V
否
检查耳机 HDC802、HDC803，
更换 U700 或更换主板
是
检查 R819 的时钟脉冲是否为 24MHz
否
检查与U601相关的脉冲是否为
24MHz
是
检查 R818 的时钟脉冲是否为
54MHz
否
检查HDC802的管脚21
（管脚19、20是否发生短路）
是
更换照相机模块
是
结束


CAM_3M_1.2V
CAM_IO_2.8V
CAM_AF_3.0V
CAM_A_2.8V
HDC802
L802
CAM_MEGA_nRST
CAM_MEGA_EN
CAM_F(4)
CAM_F(5)
CAM_F(6)
CAM_F(7)
CAM_F(0)
CAM_F(1)
CAM_F(2)
CAM_F(3)
CAM_PCLK
R818
R819
CAM_MCLK
L803
CAM_VSYNC
CAM_HSYNC
CAM_SCL_2.8V
CAM_SDA_2.8V
D800
C815
C824
C825
C816
D803
C826
C827

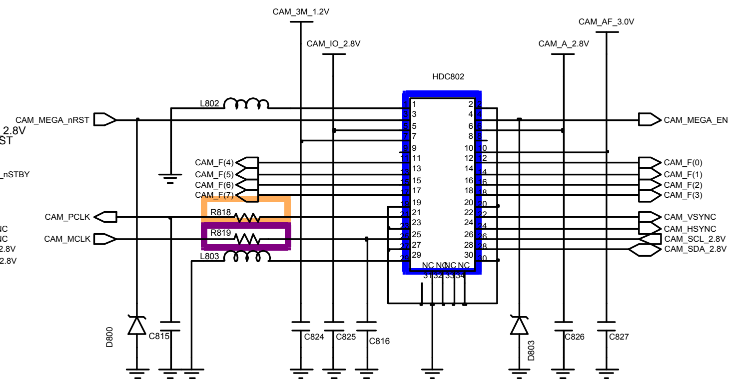

U107
HDC401
LED800
U810
L805
C854
F802
F800
F801
HDC802
L111
U102
L806
C139
BH101
BH109

8-3-11. 1.3M照相机

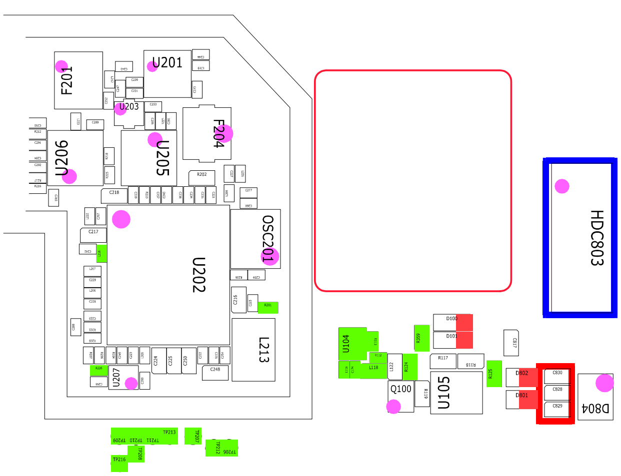

“照相机”功能不运作
是
检查主板上的照相机连接器
否
重新连接 HDC802、HDC803
是
检查电压
C828 =2.8V
C829 =1.5V
C830 =2.8V
否
检查耳机 HDC802、HDC803，
更换 U700 或更换主板
是
检查 R819 的时钟脉冲是否为 12MHz
否
检查与U601相关的脉冲是否为
12MHz
是
检查 R818 的时钟脉冲是否为
24MHz
否
检查HDC802的管脚21
（管脚19、20是否发生短路）
是
更换照相机模块
是
结束

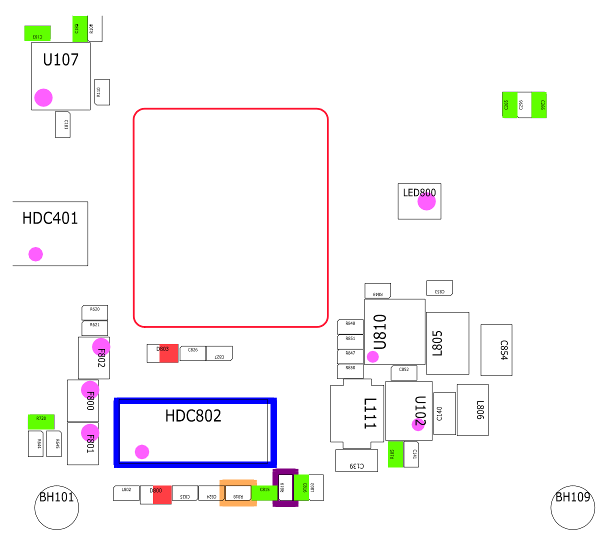


1.3M_CAM & LUME CONN.

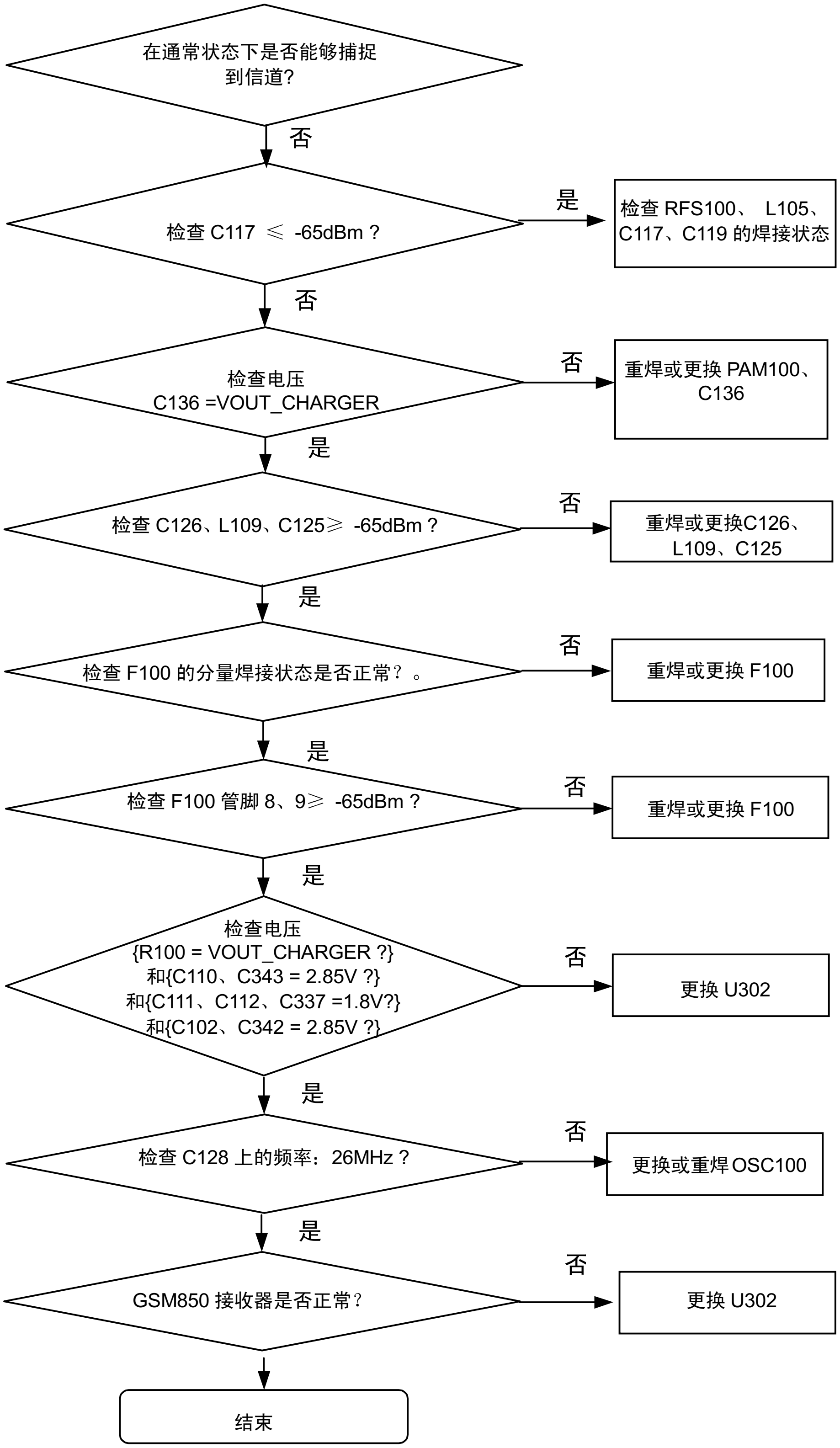


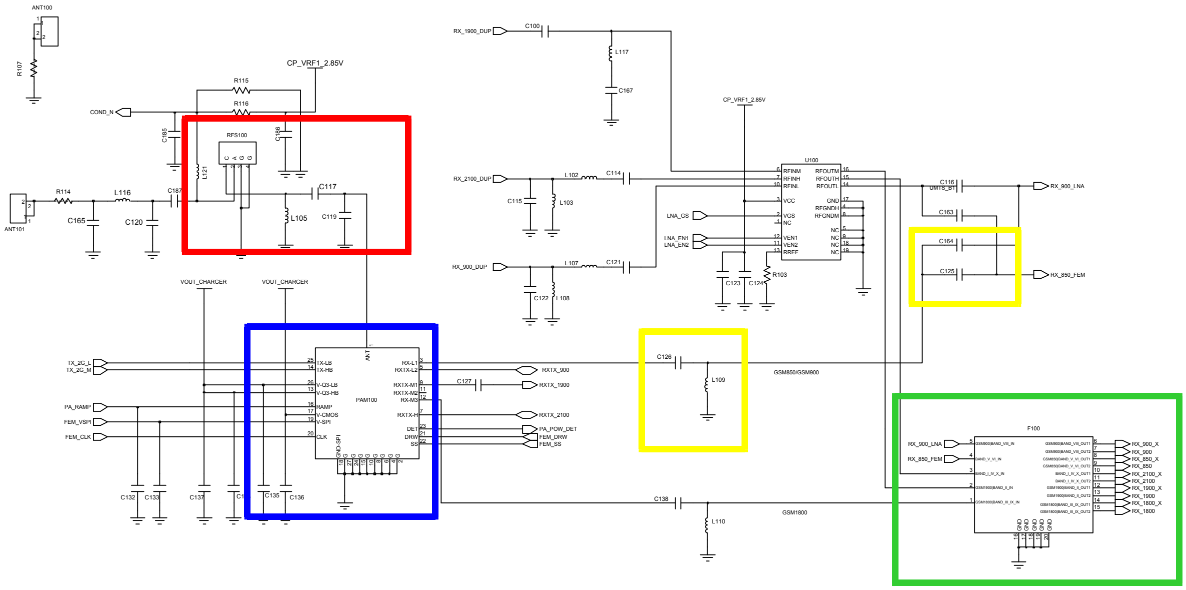
3M_CAMERA CONN.

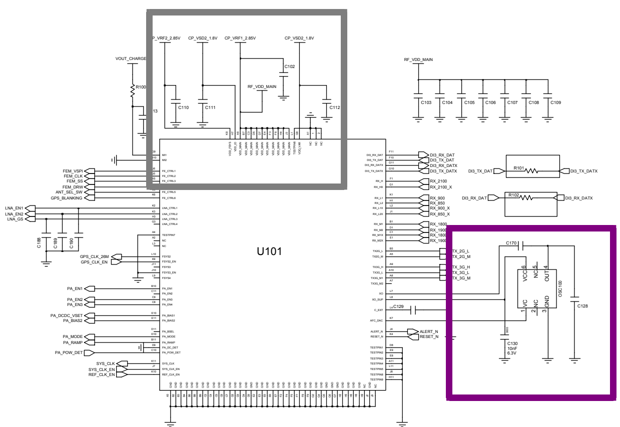


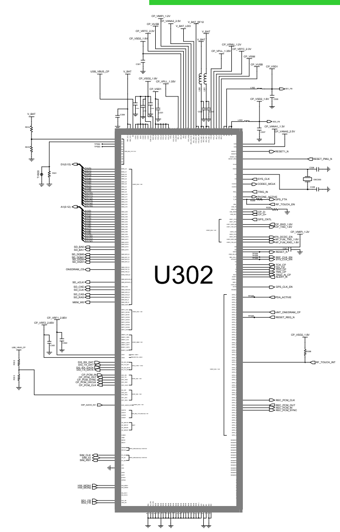


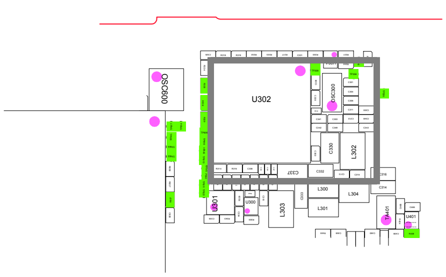
8-3-12.GSM850接收器

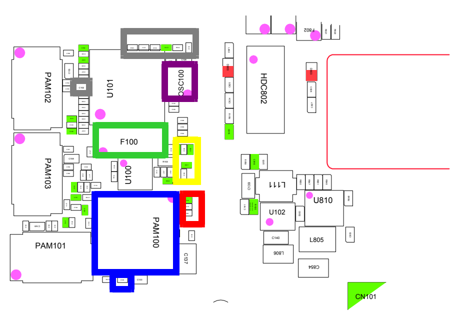



ANT100
R107
CP_VRF1_2.85V
R115
R116
COND_N
RFS100
C165
C166
L121
C117
L105
C119
R114
L116
C167
C165
C120
ANT101
VOUT_CHARGER
VOUT_CHARGER
TX_2G_L
TX_2G_M
PA_RAMP
FEM_VSPI
FEM_CLK
PAM100
C132
C133
C137
C135
C136
RX_1900_DUP
C100
L117
C167
CP_VRF1_2.85V
U100
RX_2100_DUP
L102
C114
C115
L103
LNA_GS
LNA_EN1
LNA_EN2
RX_900_DUP
L107
C121
C122
L108
C123
C124
R103
C116
C163
C164
C125
RX_900_LNA
RX_850_FEM
RXTX_900
C127
RXTX_1900
RXTX_2100
PA_POW_DET
FEM_DRW
FEM_SS
C126
L109
GSM850/GSM900
F100
RX_900_LNA
RX_850_FEM
RX_900_X
RX_900
RX_850_X
RX_850
RX_2100_X
RX_2100
RX_1900_X
RX_1900
RX_1800_X
RX_1800
C138
L110
GSM1800

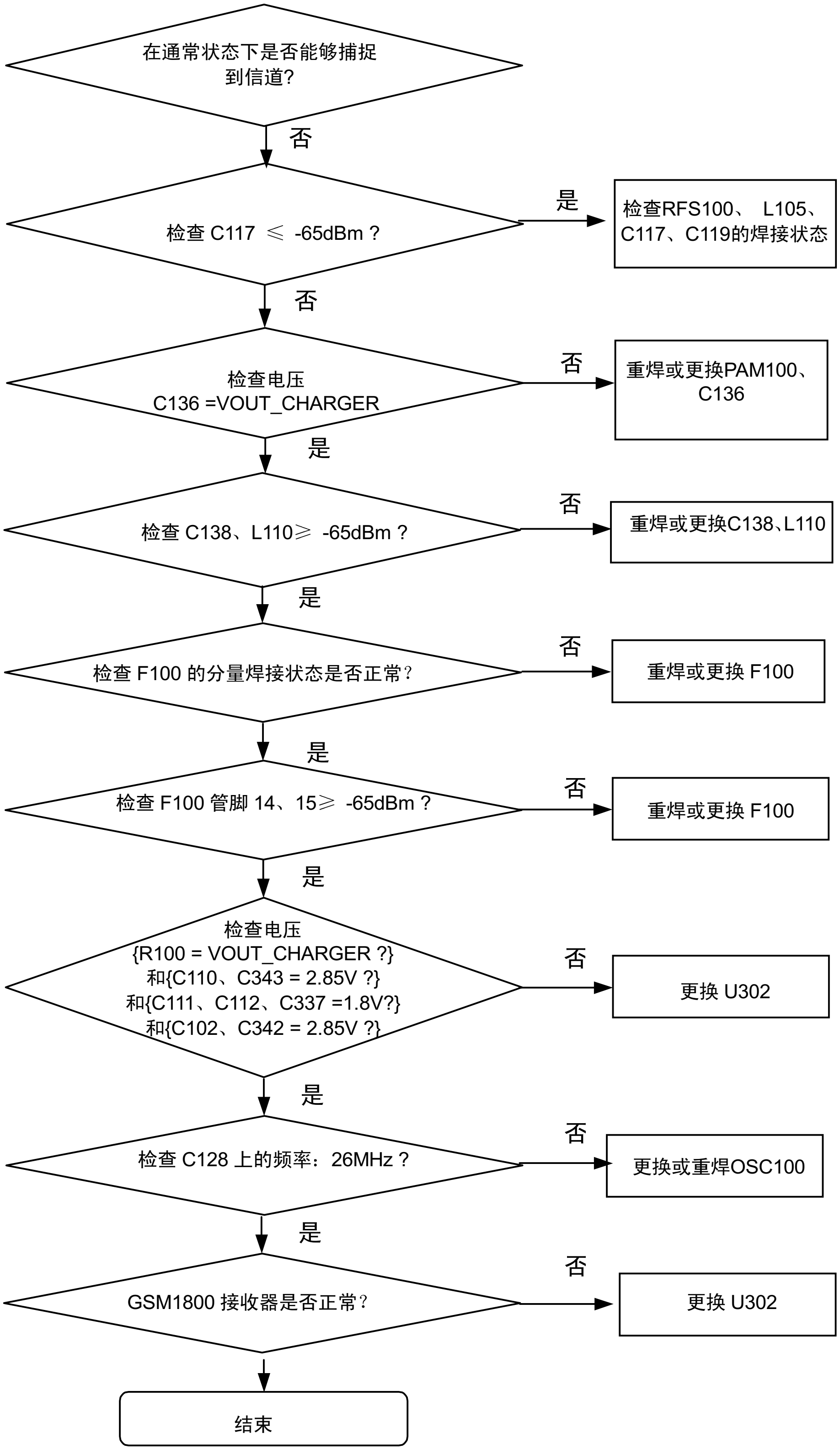

U101
C129
C130
C128
OSC100

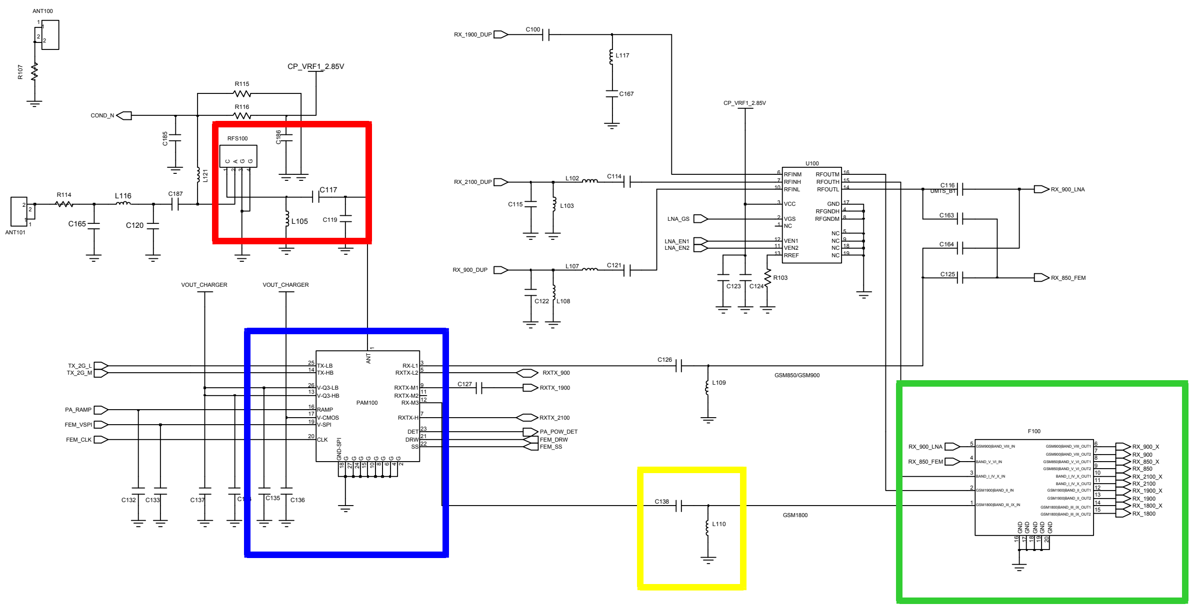

U302

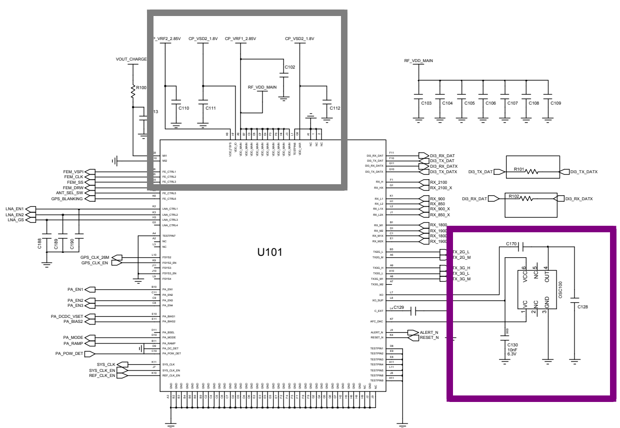

OSC300
U302
L302
L303

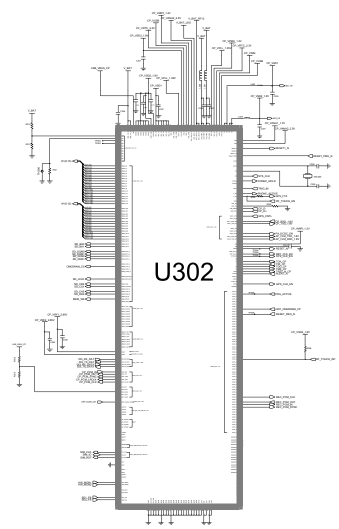

PAM102
U101
OSC100
HDC802
F100
PAM103
U100
PAM100
PAM101
U102
U810
L805
CN101

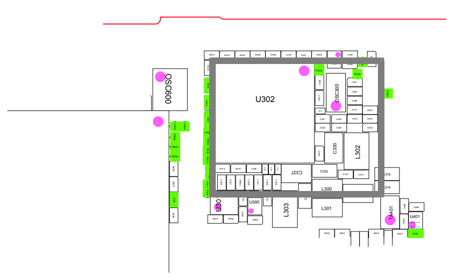
8-3-13.GSM1800接收器

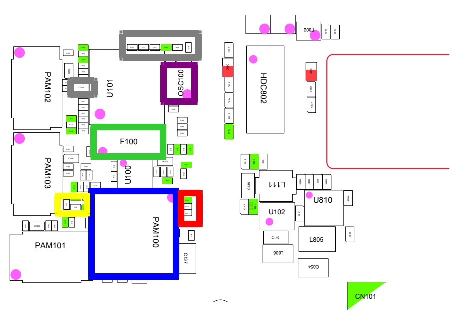



ANT100
ANT101
R107
R114
L116
C165
C120
C187
L121
C185
C186
R115
R116
COND_N
CP_VRF1_2.85V
RFS100
C117
L105
C119
VOUT_CHARGER
VOUT_CHARGER
TX_2G_L
TX_2G_M
PA_RAMP
FEM_VSPI
FEM_CLK
PAM100
C132
C133
C137
C135
C136
RX_1900_DUP
C100
L117
C167
CP_VRF1_2.85V
U100
RX_2100_DUP
L102
C114
C115
L103
LNA_GS
LNA_EN1
LNA_EN2
RX_900_DUP
L107
C121
C122
L108
C123
C124
R103
C116
C163
C164
C125
RX_900_LNA
RX_850_FEM
C126
L109
GSM850/GSM900
C127
RXTX_900
RXTX_1900
RXTX_2100
PA_POW_DET
FEM_DRW
FEM_SS
C138
L110
GSM1800
F100
RX_900_LNA
RX_850_FEM

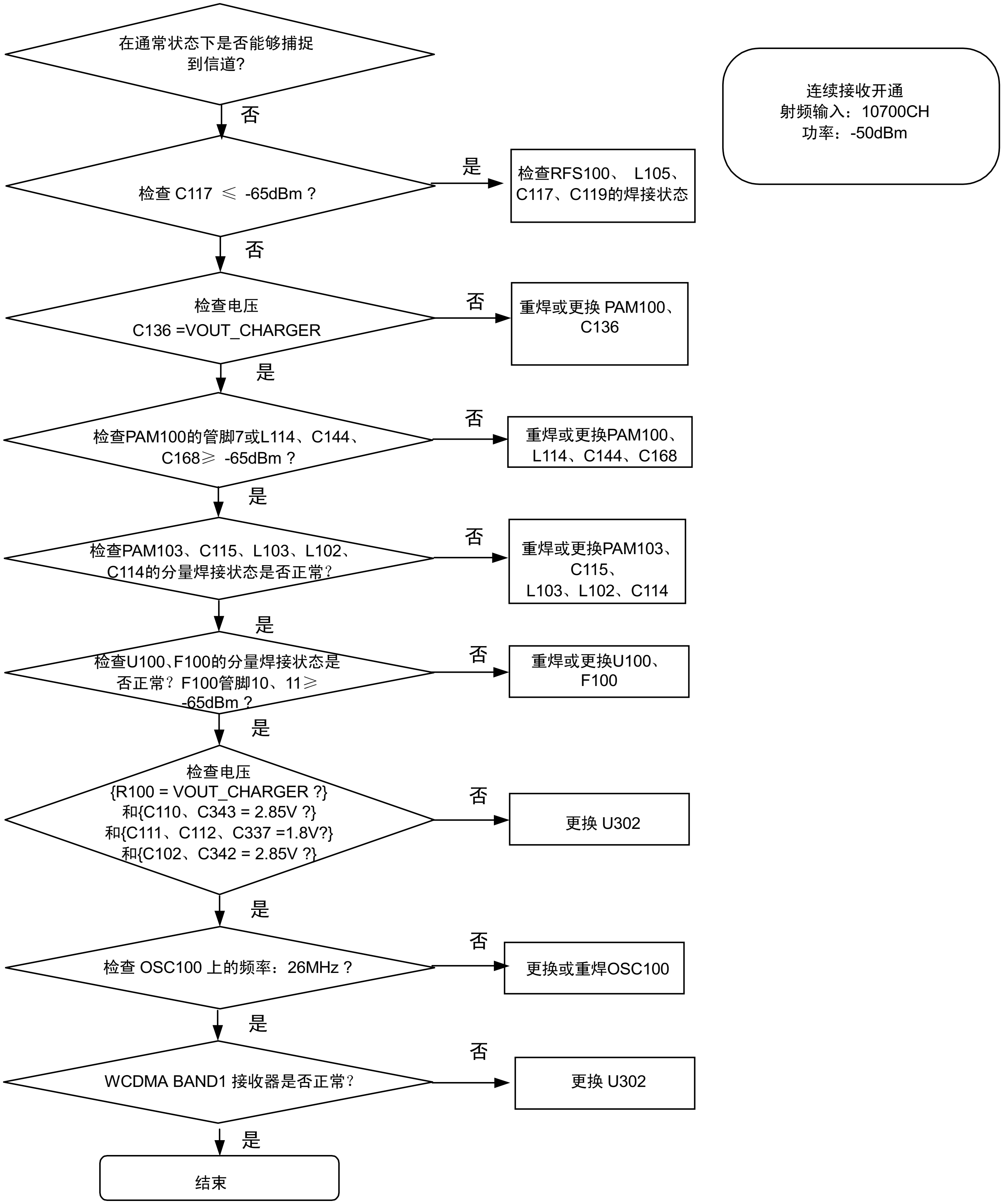

U101
VOUT_CHARGE
RF_VDD_MAIN
FEM_VSPI
FEM_CLK
FEM_SS
FEM_DRW
ANT_SEL_SW
GPS_BLANKING
LNA_EN1
LNA_EN2
LNA_GS
GPS_CLK_26M
GPS_CLK_EN
PA_EN1
PA_EN2
PA_EN3
PA_DCDC_VSET
PA_BIAS
PA_MODE
PA_RAMP
PA_POW_DET
SYS_CLK
SYS_CLK_EN
REF_CLK_EN
DG_TX_DATX
DG_RX_DATX
OSC100
ALERT_N
RESET_N

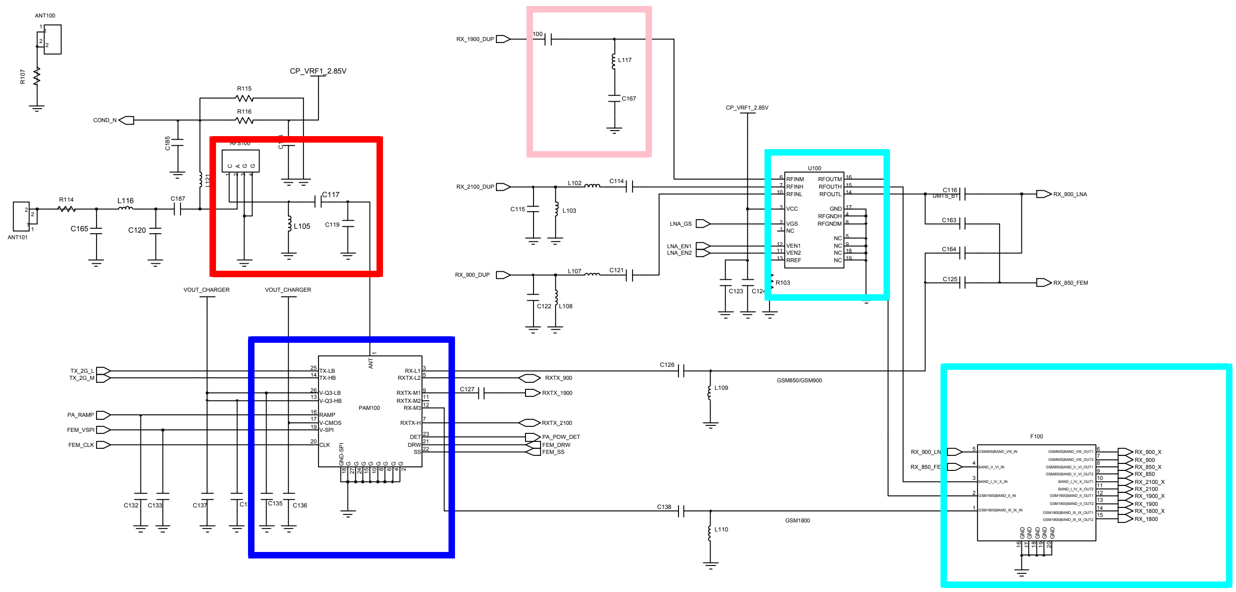

U302

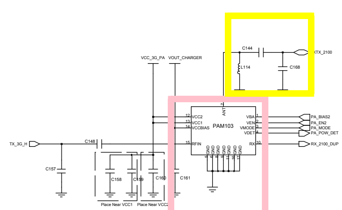

OSC600
U302
L302
L303
L300
L301

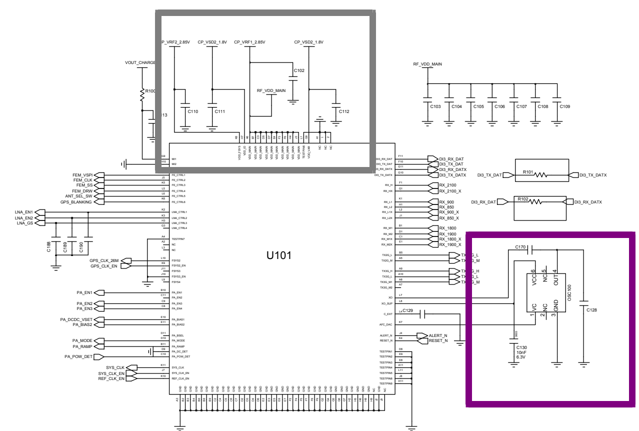

PAM102
U101
OSC100
HDC802
F100
PAM103
U100
PAM100
PAM101
L111
U810
U102
L805
CN101

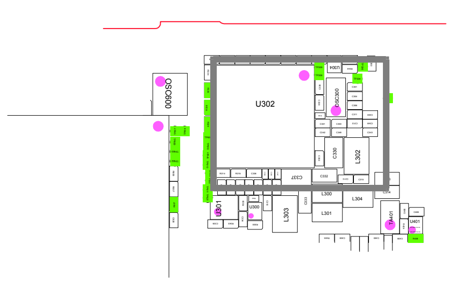
8-3-14.WCDMA Band1 接收器

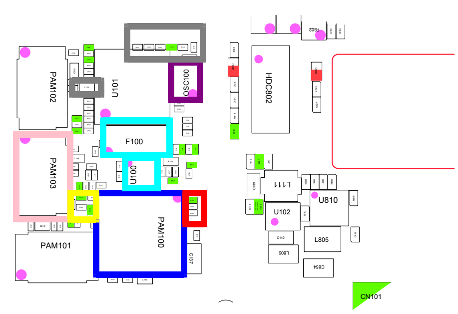



ANT100
R107
CP_VRF1_2.85V
R115
R116
COND_N
C185
L121
C117
L105
C119
R114
L116
C187
C165
C120
ANT101
VOUT_CHARGER
RX_1900_DUP
L117
C167
RX_2100_DUP
L102
C114
C115
L103
RX_900_DUP
L107
C121
C122
L108
CP_VRF1_2.85V
U100
LNA_GS
LNA_EN1
LNA_EN2
R103
C123
C124
C116
C163
C164
C125
RX_900_LNA
RX_850_FEM
TX_2G_L
TX_2G_M
PA_RAMP
FEM_VSPI
FEM_CLK
PAM100
C126
L109
GSM850/GSM900
C127
RXTX_900
RXTX_1900
RXTX_2100
PA_POW_DET
FEM_DRW
FEM_SS
C132
C133
C137
C135
C136
C138
L110
GSM1800
F100
RX_900_LNA
RX_850_FEM
RX_900_X
RX_900
RX_850_X
RX_850
RX_2100_X
RX_2100
RX_1900_X
RX_1900
RX_1800_X
RX_1800

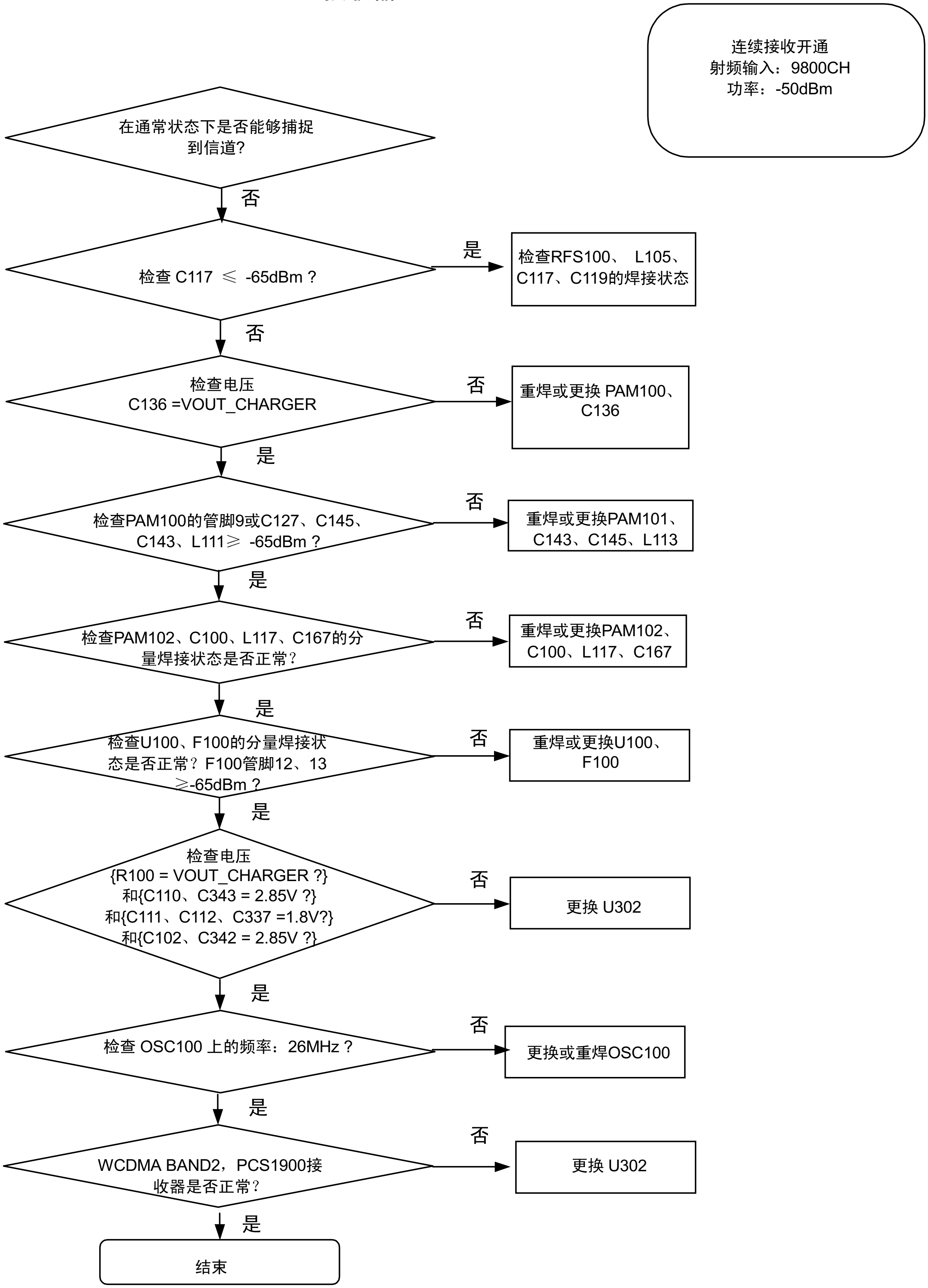

C144
L114
C188
TX_2100
VCC_3G_PA
VOUT_CHARGER
PAM103
TX_3G_H
C148
C197
PA_BIAS2
PA_EN2
PA_MODE
PA_POW_DET
RX_2100_DUP

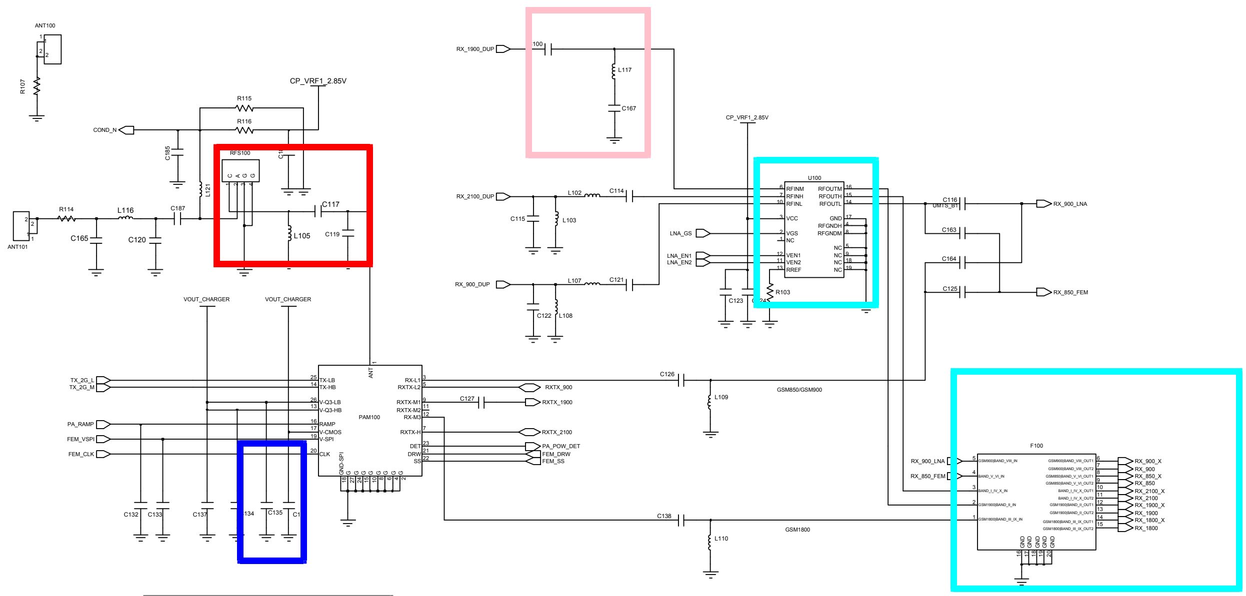

VOUT_CHARGER
CP_VRF2_2.85V
CP_VDD2_1.8V
CP_VRF1_2.85V
CP_VDD2_1.8V
RF_VDD_MAIN
C110
C111
C112
RF_VDD_MAIN
C103
C104
C105
C106
C107
C108
C109
U101
DIG_TX_DATX
DIG_RX_DATX
C170
OSC100
C128
C129
C130
ALERT_N
RESET_N
GPS_CLK_26M
GPS_CLK_EN
SYS_CLK
SYS_CLK_EN
REF_CLK_EN

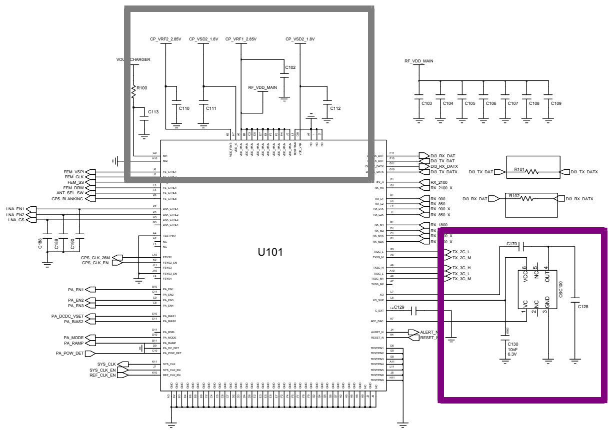

OSC600
U302
C337
C330
L302
L300
L304
L301
L303

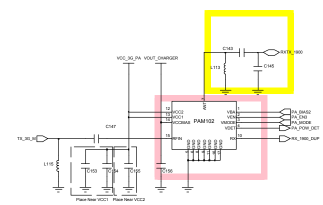

PAM102
U101
OSC100
HDC802
F100
U100
PAM103
PAM100
PAM101
C137
L111
U102
U810
L805
C804
CN101

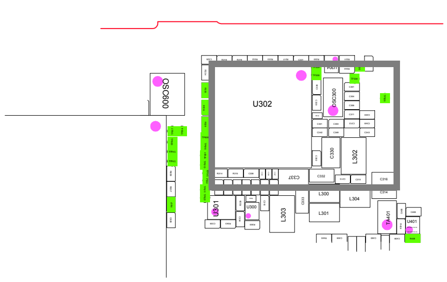
8-3-15.WCDMA Band2/GSM1900接收器




ANT100
ANT101
RFS100
PAM100
U100
F100
CP_VRF1_2.85V
VOUT_CHARGER

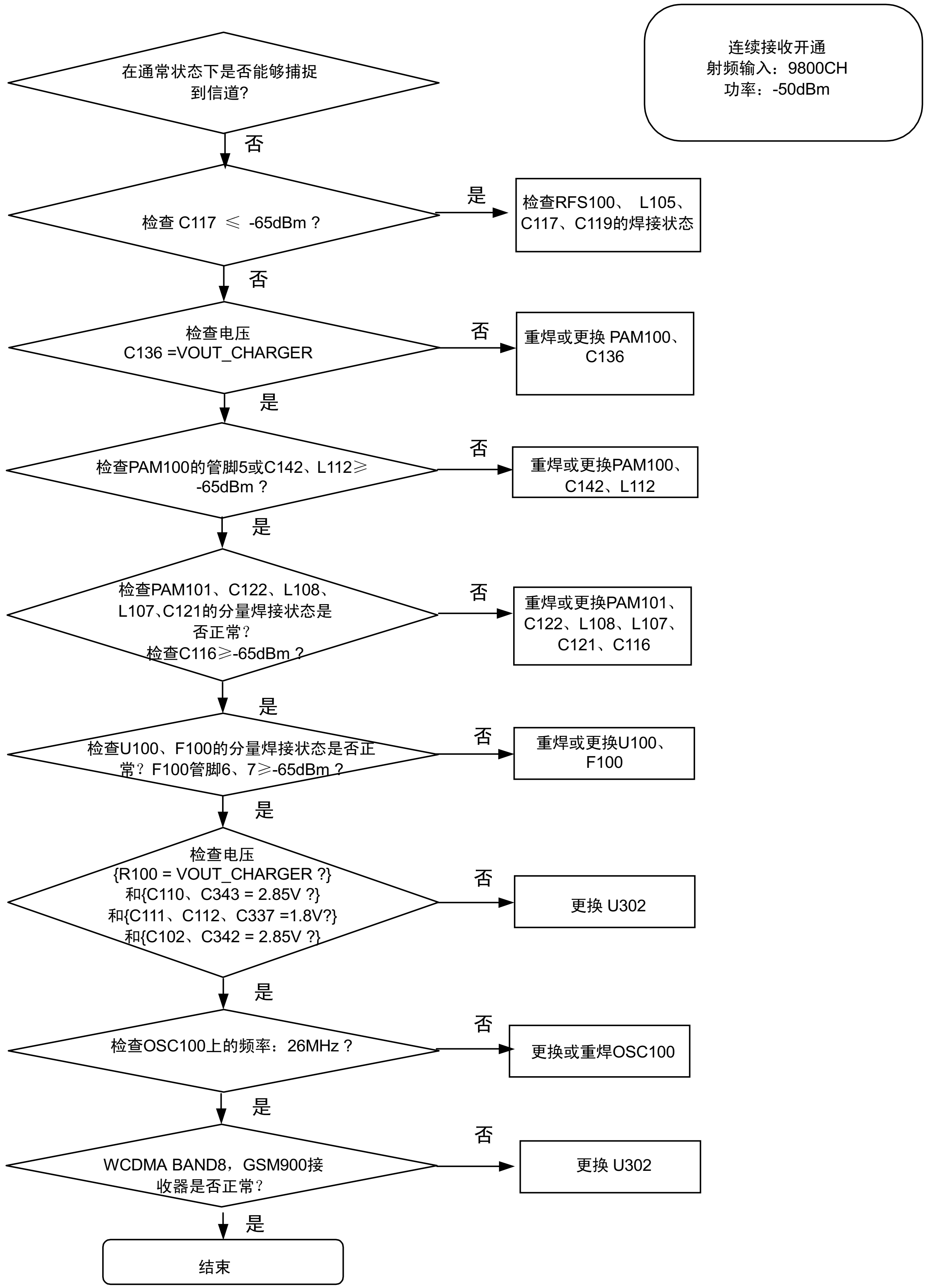

U101

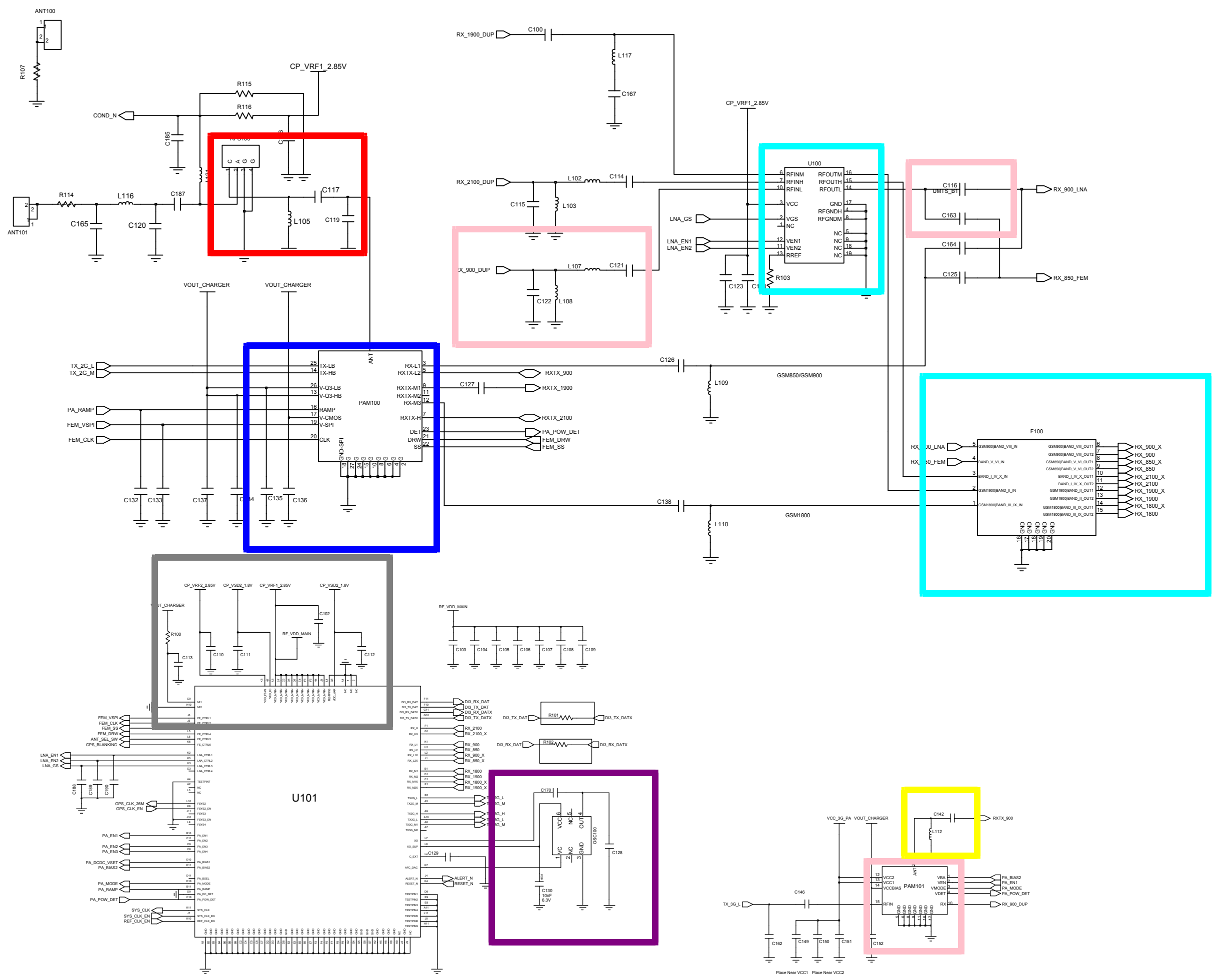

PAM102

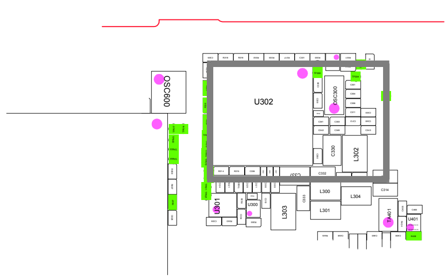

U302
OSC600

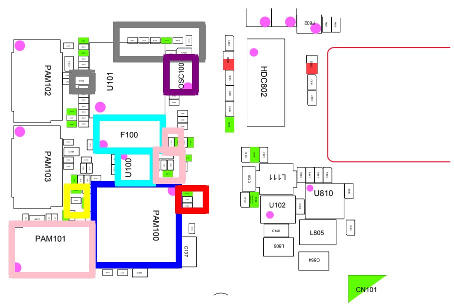

PAM102
U101
OSC100
HDC802
F100
U100
PAM103
PAM100
U810
U102
L805
PAM101
CN101

8-3-16.WCDMA Band8/GSM900接收器



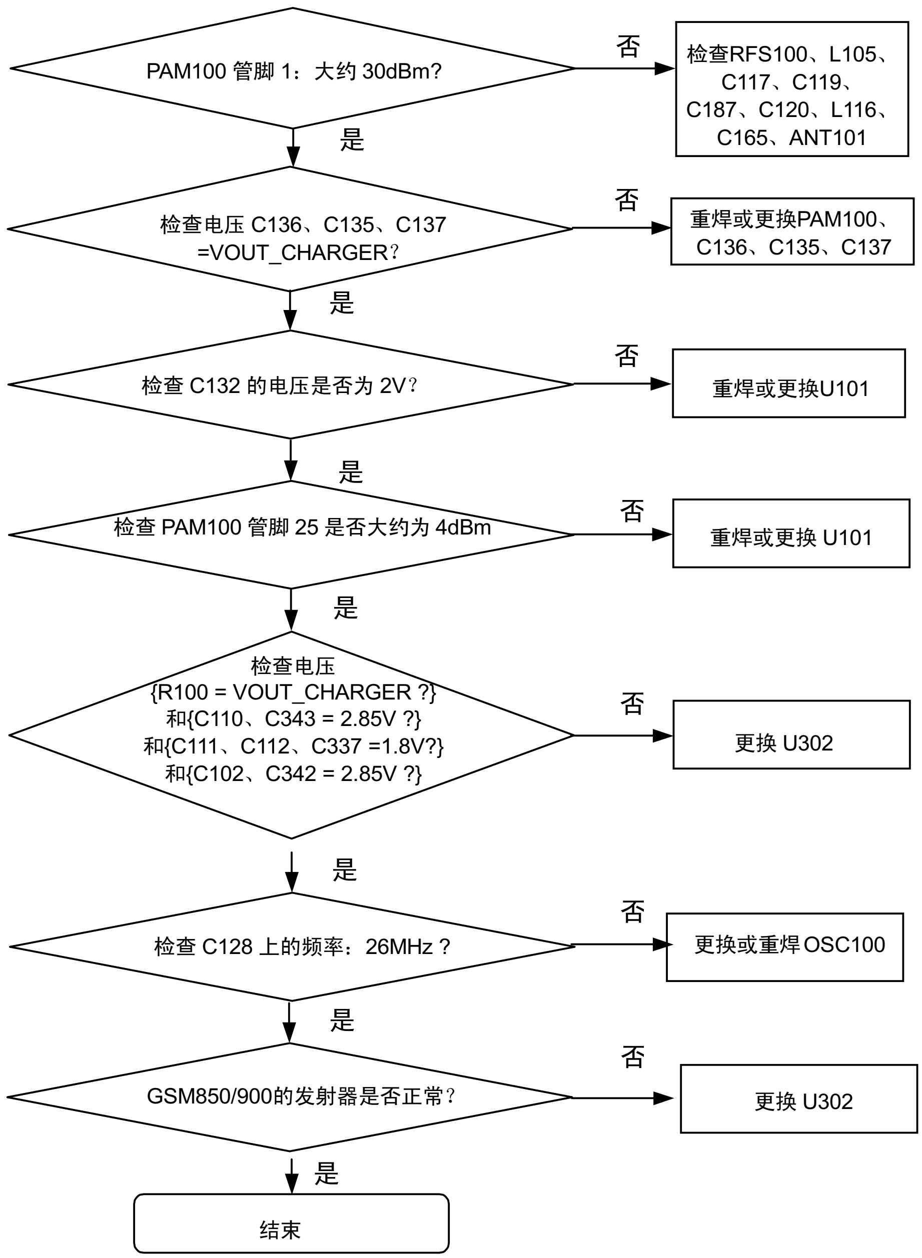




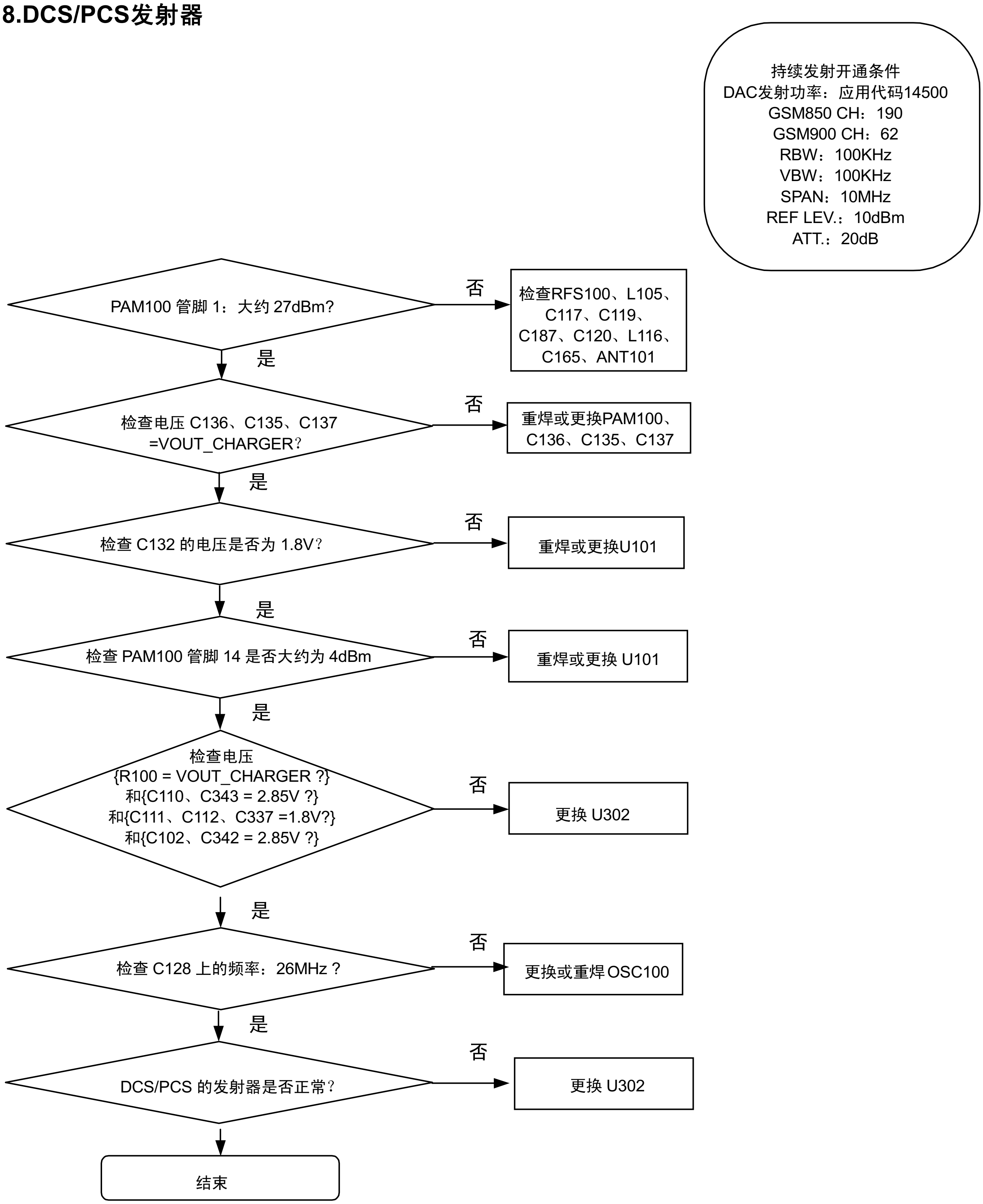


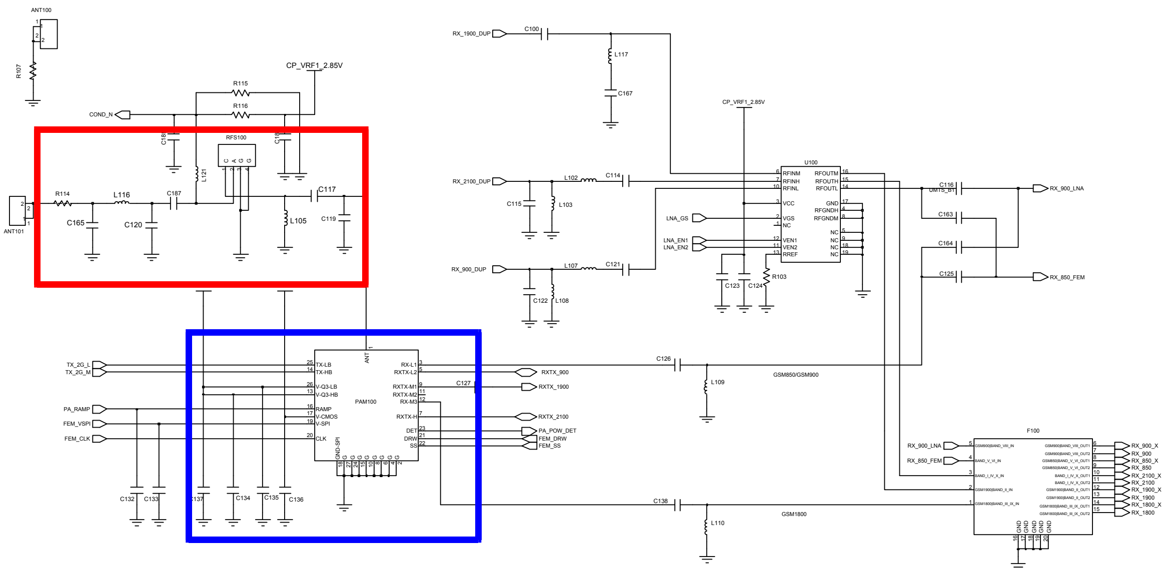
8-3-17.GSM850/GSM900发射器

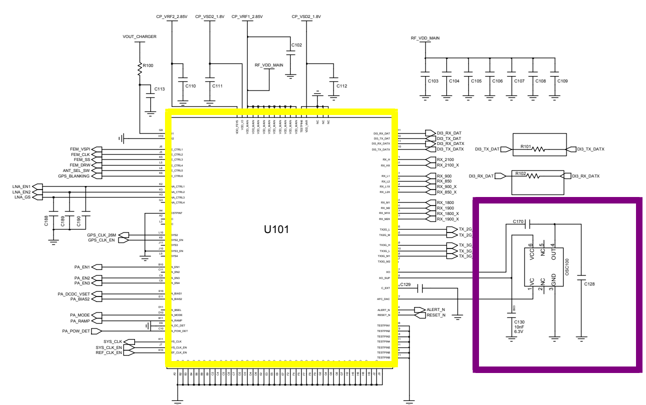
持续发射开通条件
DAC发射功率：应用代码14500
GSM850 CH：190
GSM900 CH：62
RBW：100KHz
VBW：100KHz
SPAN：10MHz
REF LEV.：10dBm
ATT.：20dB



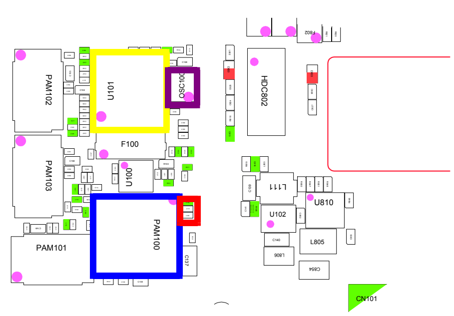
8-3-18.DCS/PCS发射器



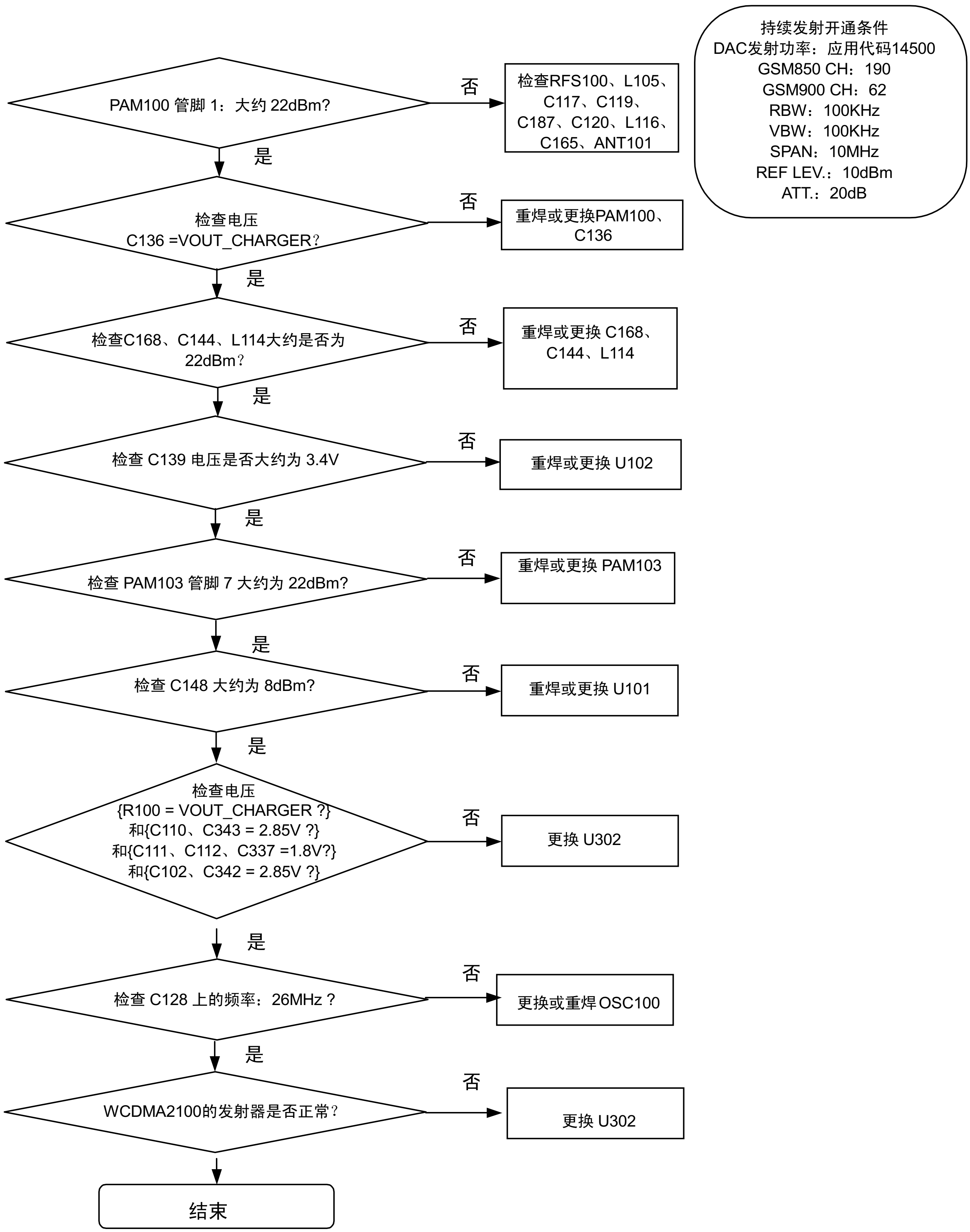

ANT100
ANT101
CP_VRF1_2.85V
RFS100
PAM100
U100
F100
GSM850/GSM900
GSM1800
RX_900_LNA
RX_850_FEM
RXTX_900
RXTX_1900
RXTX_2100
PA_POW_DET
FEM_DRW
FEM_SS
TX_2G_L
TX_2G_M
PA_RAMP
FEM_VSPI
FEM_CLK

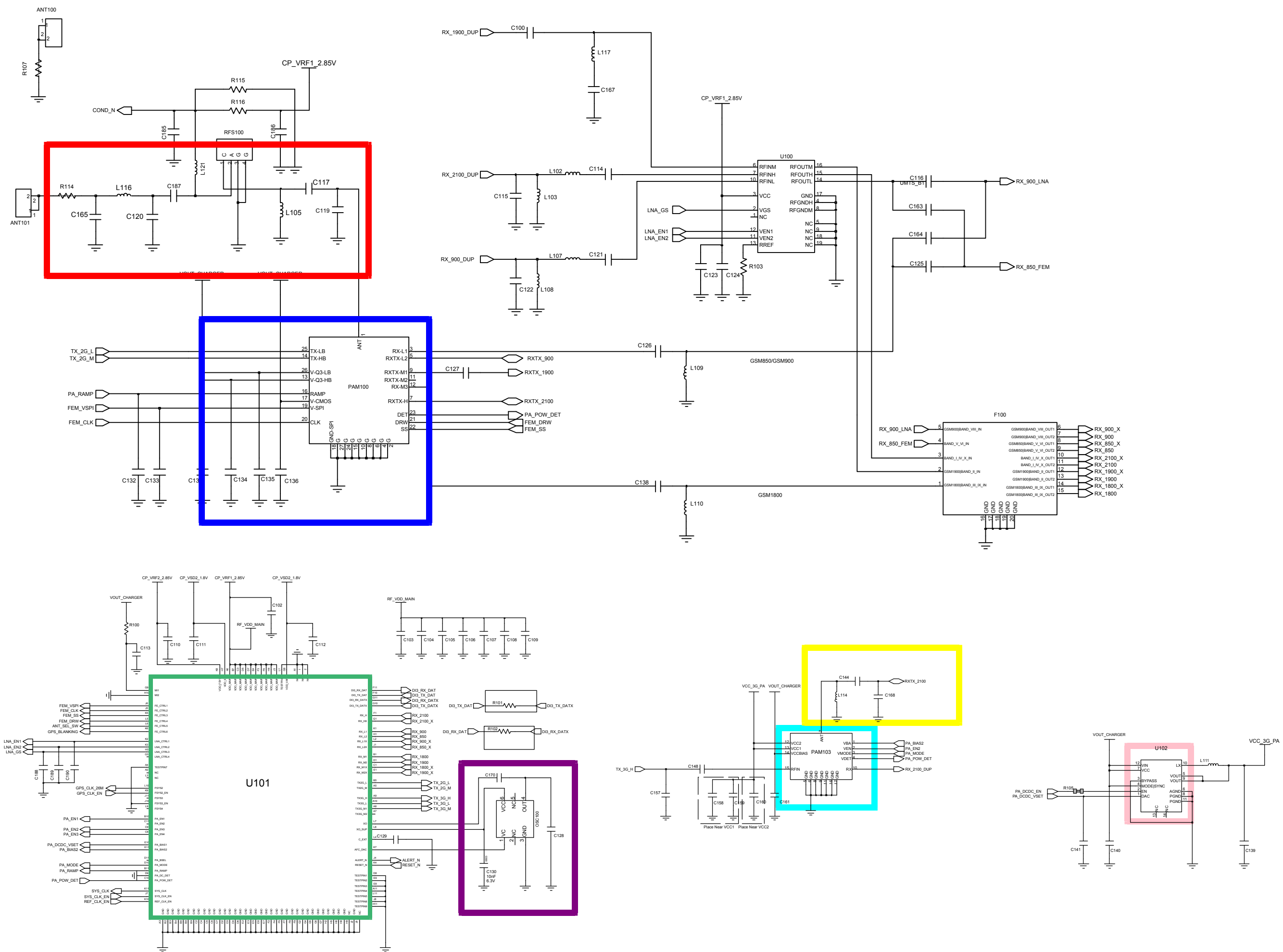

U101


U302
OSC600

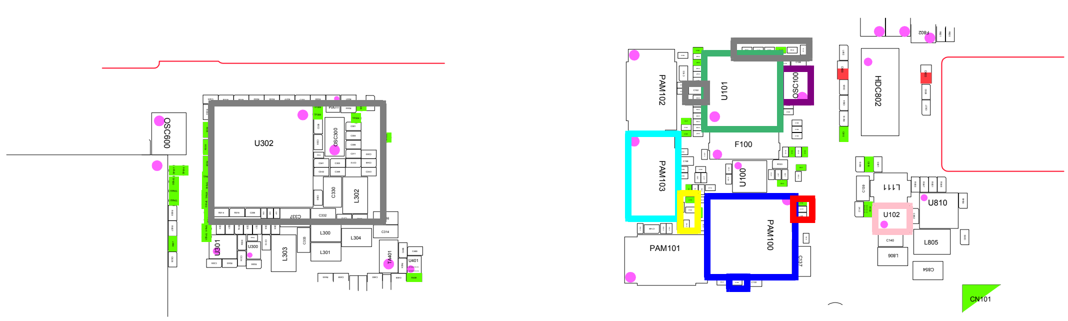

PAM102
U101
PAM103
F100
PAM101
PAM100
HDC802
U810
U102
L805
CN101

8-3-19.WCDMA BAND1发射器



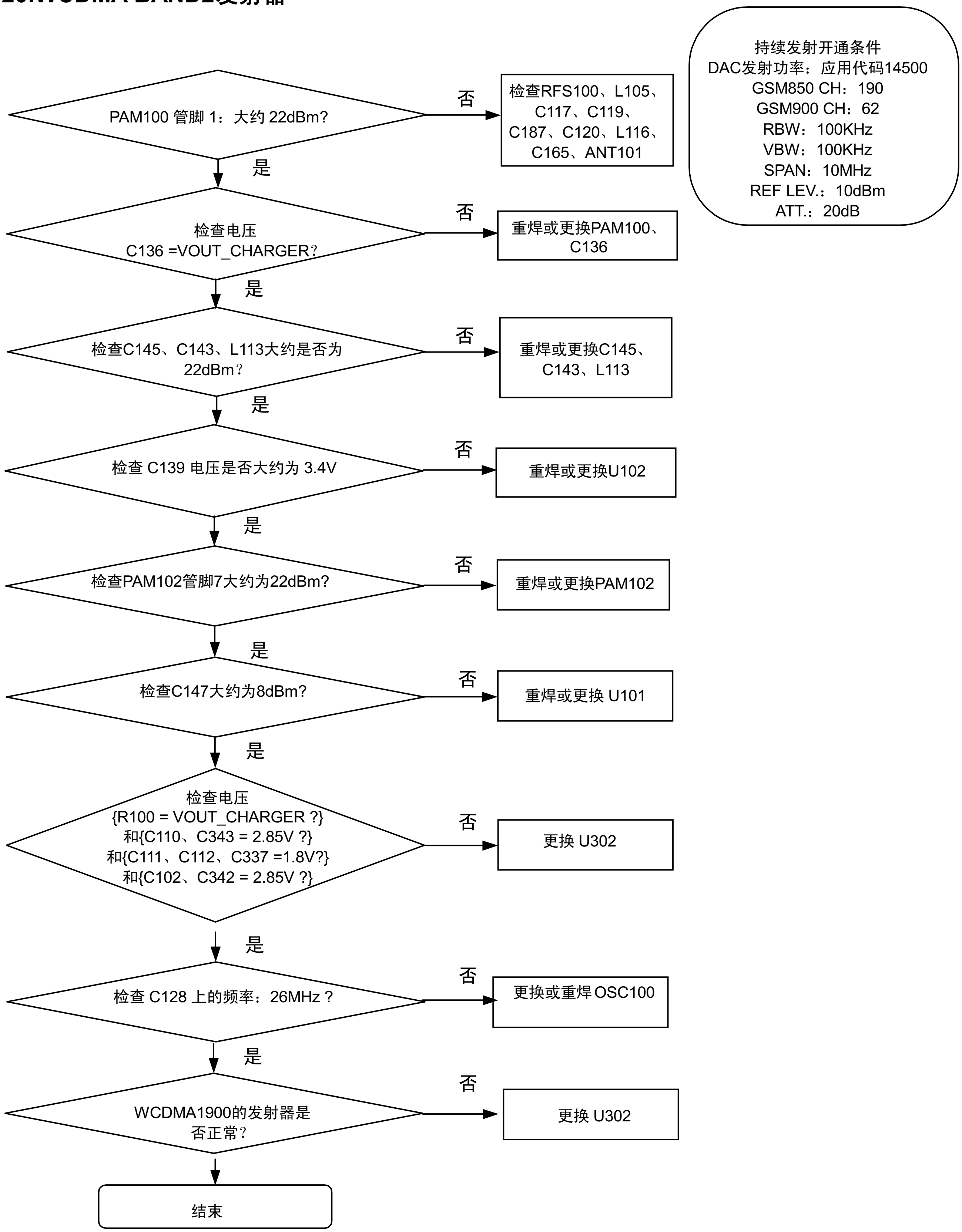


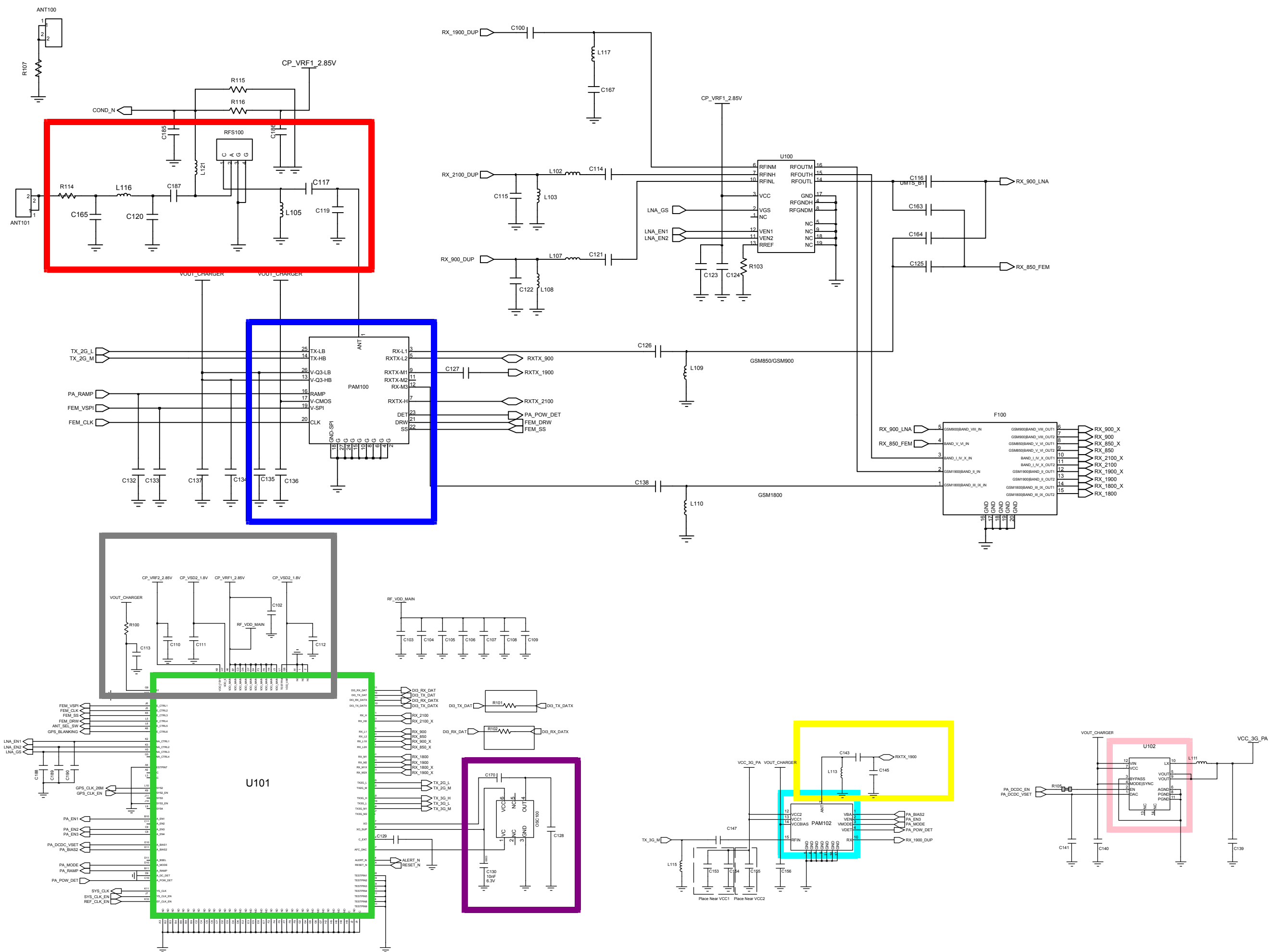
ANT100
R107
COND_N
CP_VRF1_2.85V
R115
R116
RFS100
ANT101
R114
L116
C187
C117
C165
C120
L105
C119
L121
C185
C186
TX_2G_L
TX_2G_M
PA_RAMP
FEM_VSPI
FEM_CLK
PAM100
C132
C133
C134
C135
C136
RX_1900_DUP
C100
L117
C167
RX_2100_DUP
L102
C114
C115
L103
RX_900_DUP
L107
C121
C122
L108
U100
LNA_GS
LNA_EN1
LNA_EN2
R103
C123
C124
C116
C163
C164
C125
RX_900_LNA
RX_850_FEM
C126
L109
GSM850/GSM900
C127
RXTX_900
RXTX_1900
RXTX_2100
PA_POW_DET
FEM_DRW
FEM_SS
F100
C138
L110
GSM1800
U101



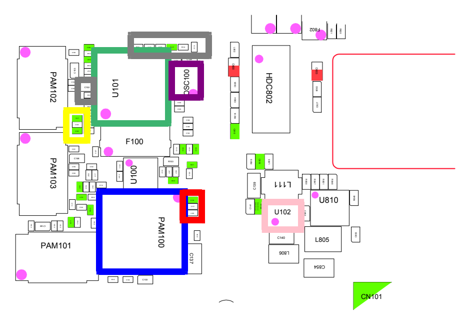
OSC600
U302
PAM102
U101
F100
PAM103
PAM101
PAM100
HDC802
U102
U810
L805
CN101

8-3-20.WCDMA BAND2发射器

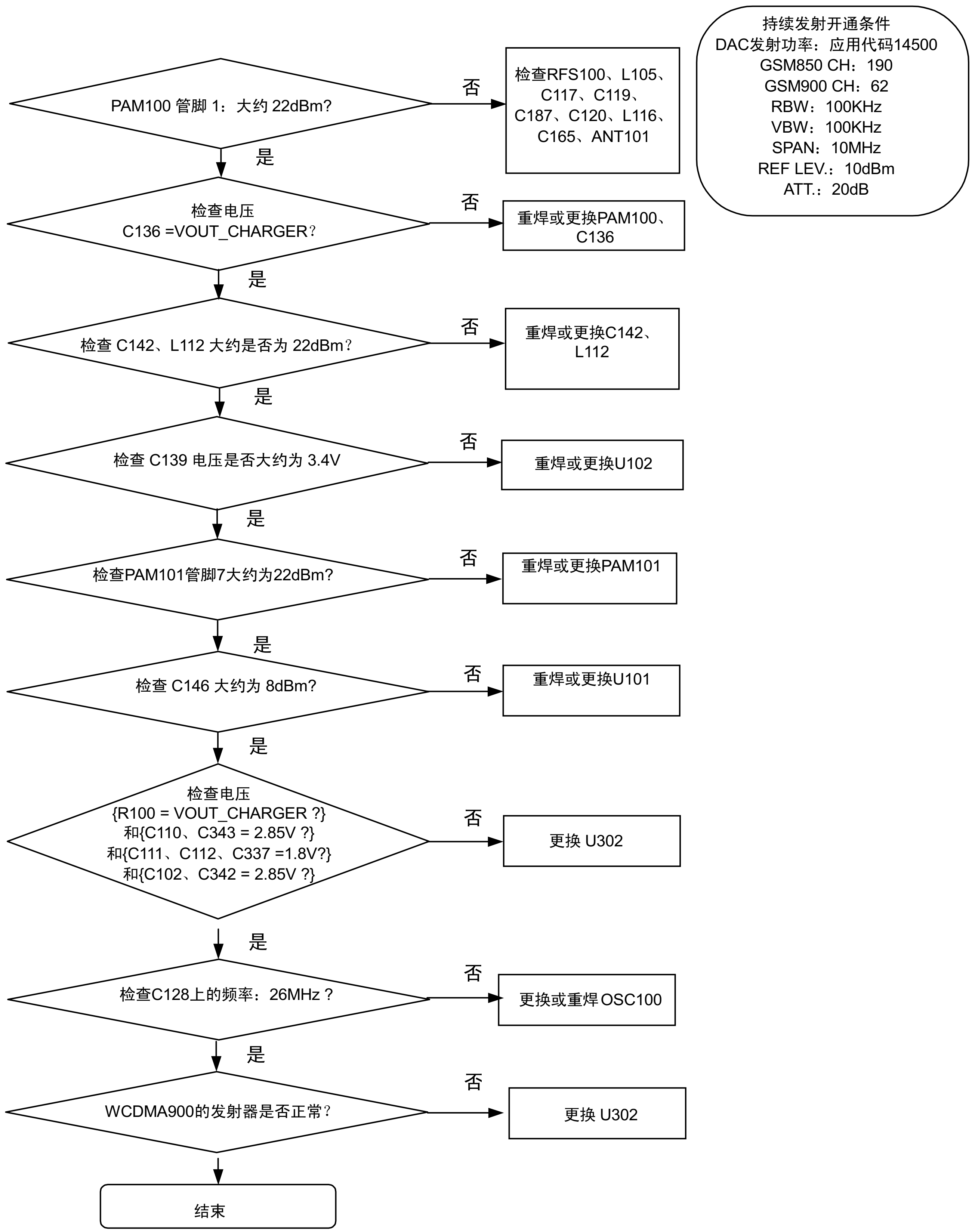


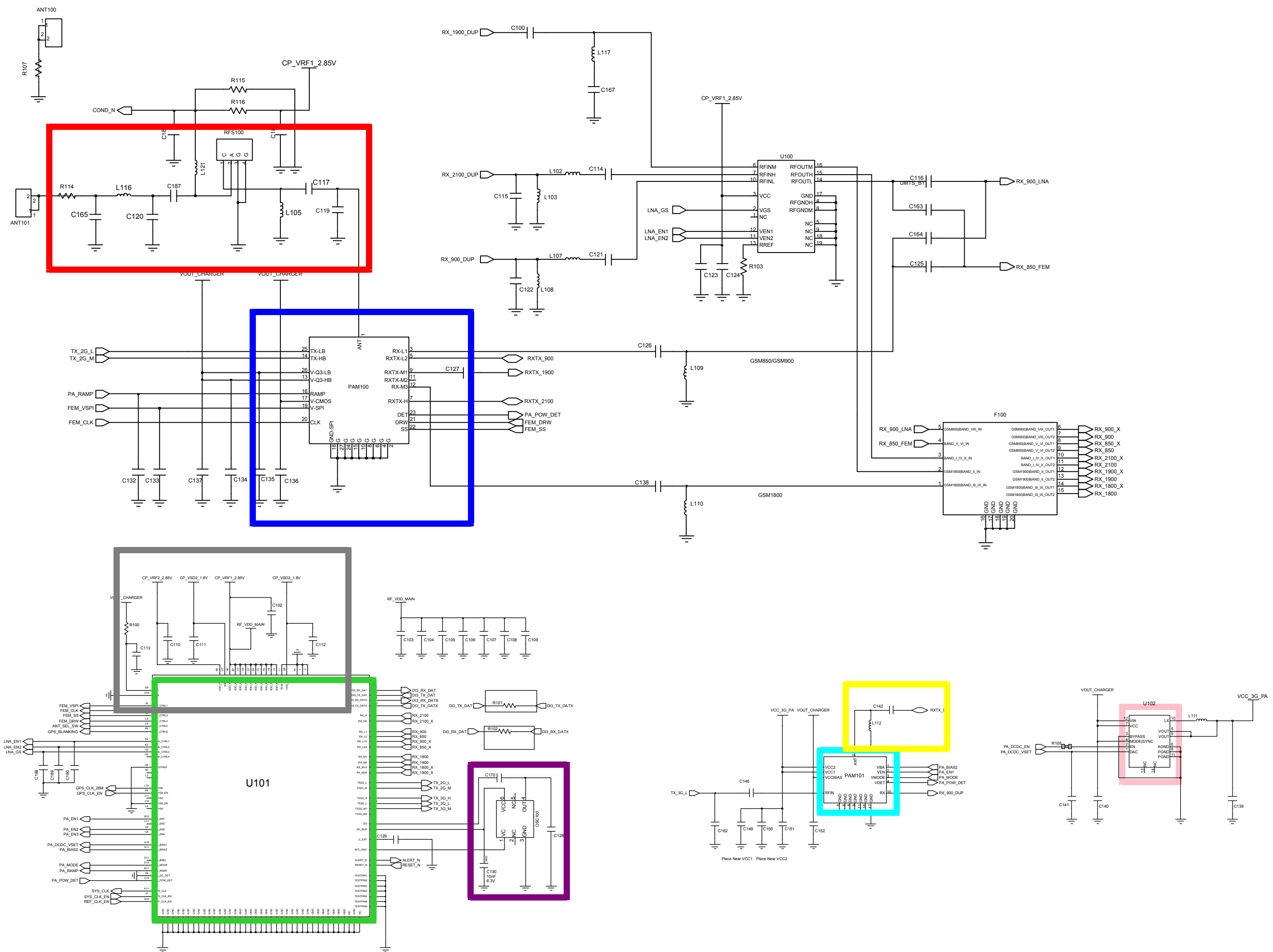

ANT100
R107
CP_VRF1_2.85V
R115
R116
COND_N
RFS100
ANT101
R114
L116
C187
C117
C165
C120
L105
C119
L121
RX_1900_DUP
C100
L117
C167
CP_VRF1_2.85V
U100
RX_2100_DUP
L102
C114
C115
L103
LNA_GS
LNA_EN1
LNA_EN2
RX_900_DUP
L107
C121
C122
L108
C123
C124
R103
C116
C163
C164
C125
RX_900_LNA
RX_850_FEM
TX_2G_L
TX_2G_M
PA_RAMP
FEM_VSPI
FEM_CLK
PAM100
C126
L109
GSM850/GSM900
C127
RXTX_900
RXTX_1900
RXTX_2100
PA_POW_DET
FEM_DRW
FEM_SS
C132
C133
C137
C134
C135
C136
C138
L110
GSM1800
F100
RX_900_X
RX_900
RX_850_X
RX_850
RX_2100_X
RX_2100
RX_1900_X
RX_1900
RX_1800_X
RX_1800
U101
U102
VCC_3G_PA
PAM102
TX_3G_M
L115
C147
C143
C145
RXTX_1900
C128
C130

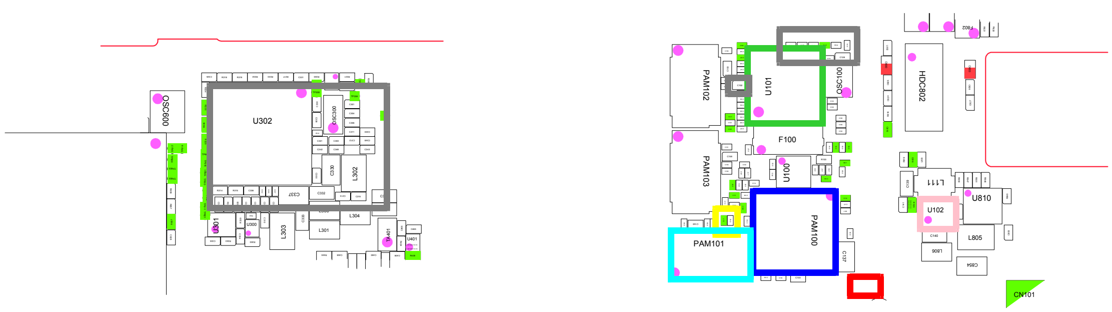

OSC600
U302
L302
L303
L300
L301
L304


PAM102
U101
OSC100
HDC802
F100
PAM103
U100
PAM100
PAM101
L111
U810
U102
L805
CN101

8-3-21.WCDMA BAND8发射器



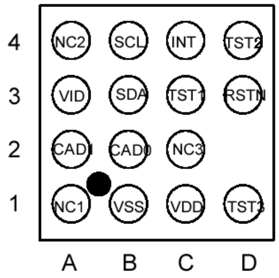




8-4.维修原理图

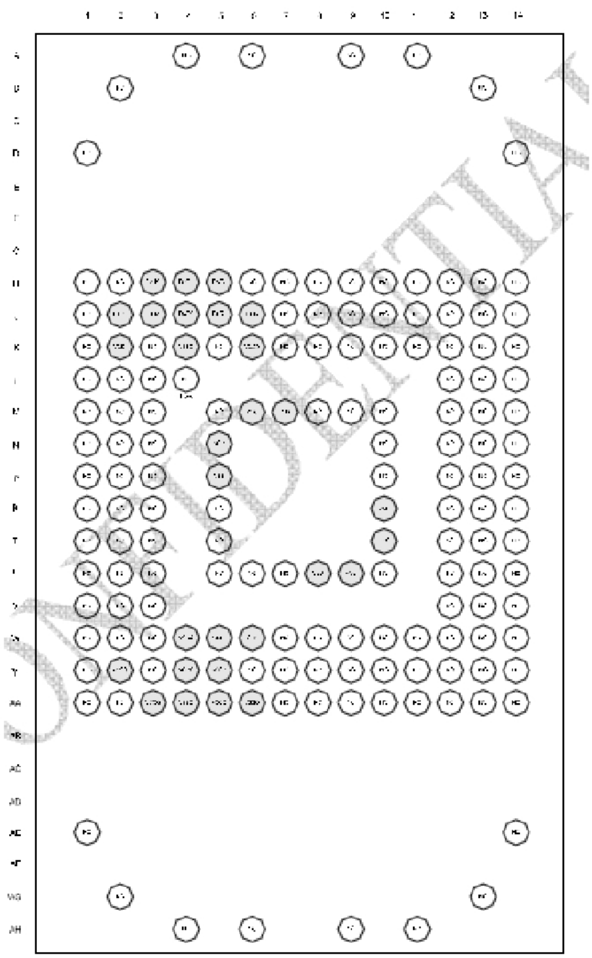
-NC 点（顶部视图）

U806



UME300

○ : **NC**



U700

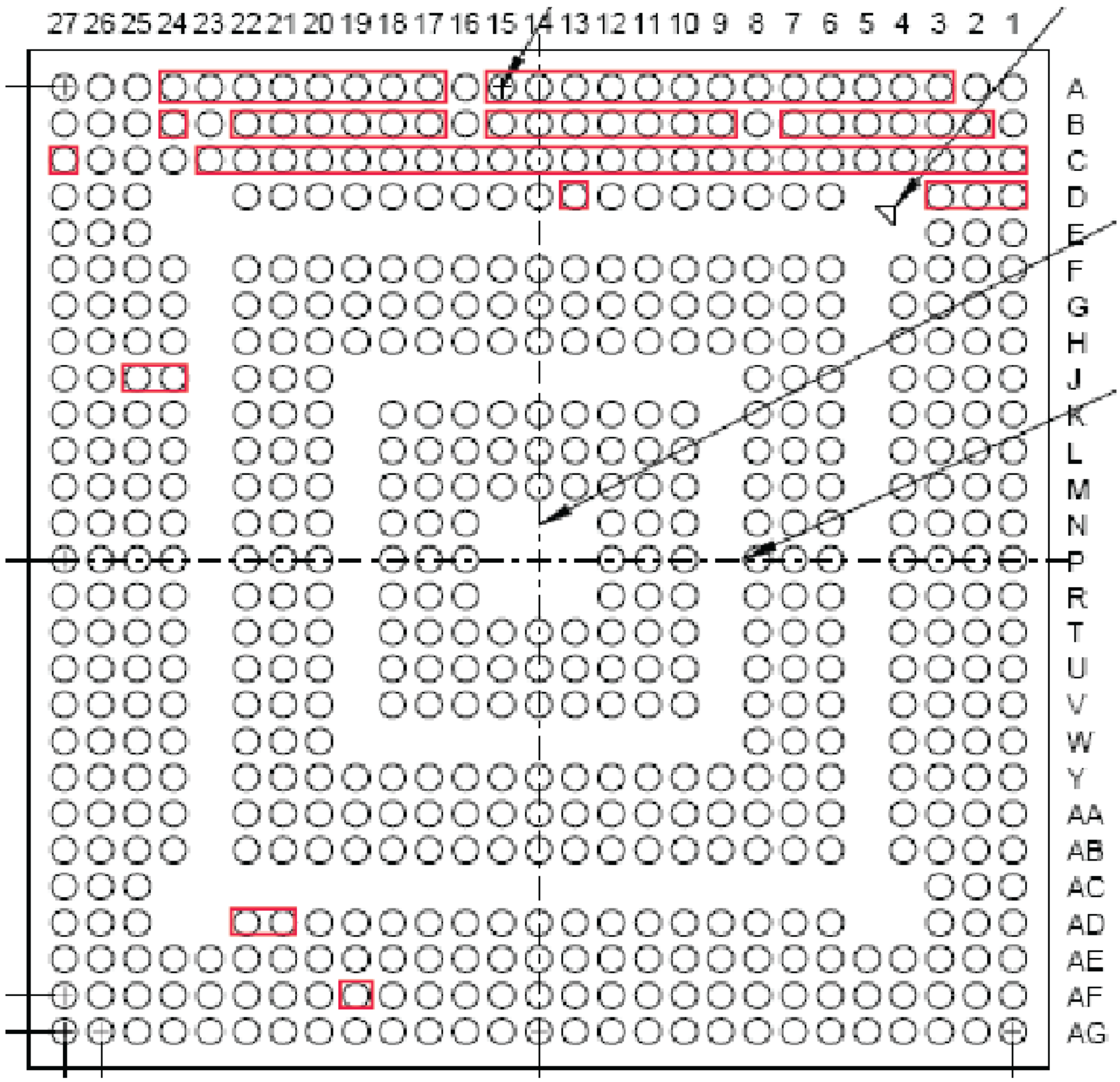
	1	2	3	4	5	6	7	8	9	10
A	IN1	IN1	XOUT	XIN	AGND 1	LDO 14	IN7	LDO 15	REF BP	LDO 16
B	LX1	LX1	32kHz CP	32kHz AP	AGND 2	VCOIN	IN7	LDO 2	IN9	LDO 17
C	LX1	BUCK 1	PWR ON	SDA	SCL	SRAD	LDO 10	LDO 3	IN6	LDO 13
D	PGND 1	PGND 1	SET1	SET2	SET3	PWR HOLD	NO22	NO21	IN6	LDO 12
E	IN2	IN2	BUCK 2	JIG ON	RSO\	PWR EN	COM 22	COM 21	VLSW 2	LDO 11
F	LX2	LX2	LX2	MR\	IRQ\	BATT MON	NC22	NC21	CS2	LDO 9
G	PGND 2	PGND 2	BUCK 3	VL	ONO\	VICHG	NO12	NO11	IN8	LDO 8
H	IN3	IN3	BUCK 4	SAFE OUT1	SAFE OUT2	DET BAT	COM 12	COM 11	IN8	LDO 6
J	LX3	LX3	LX4	PGND 4	BATT	DCIN	NC12	NC11	CS1	LDO 5
K	PGND 3	IN4	LX4	PGND 4	BATT	DCIN	LDO 7	IN5	IN5	LDO 4

U302

	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y	
20	NC	VSD2	VSD2	VSS_SD2	SD2_FB	RESET_PMU_N	EPP1	VSS_MAIN	HF_SPK1P	VBAT_HF_SPK	HF_SPK2P	CP1	MEM_AD[0]	MEM_A[0]	MEM_A[1]	MEM_A[3]	MEM_BC2_n	MEM_A[5]	MEM_A[7]	NC	20
19	VSD1	VSS_SD1	VSS_SD1	VSS_SD2	PMUMEAS	VDD_VSIM	EPN1	VSS_HF_SPK	VBAT_HF_SPK	HF_SPK1N	HF_SPK2N	CP2	MEM_AD[1]	MEM_AD[2]	MEM_A[2]	MEM_A[4]	MEM_BC0_n	MEM_A[6]	MEM_A[9]	MEM_A[8]	19
18	VDD_VBAT_SD1	VSD1				VANA2			VSS_HF_SPK			VDD_CPN			MEM_AD[6]				MEM_A[11]	MEM_A[10]	18
17	SD1_FB	VDD_VBAT_SD1		VDD_VBAT_SD2	VDD_VBAT_SD2	VDD_VPMU	VDD_VRTC	MICP1	AUX1P	AUX2P	EPPA2	EPPA1	MEM_AD[3]	MEM_AD[5]	MEM_AD[8]	MEM_AD[11]	VDDP_DIG		MEM_A[12]	MEM_BC3_n	17
16	VRF1	VSS_ANA		VSS_MAIN	VDD_VBAT_LDO	VMIPI	VANA1	MICN1	AUX1N	AUX2N	VMIC	VUMIC	MEM_AD[4]	MEM_AD[7]	MEM_AD[10]	VSS_MAIN	MEM_AD[15]		MEM_A[13]	MEM_A[15]	16
15	VREF	VDD_VBAT_RF12	VSS_RF12	VRF2	VDD_VUSB				M2			M3				VDD_CORE	MEM_WAIT_n	MEM_ADV_n	MEM_BC1_n	MEM_A[14]	15
14	Di3_TX_DAT	Di3_RX_DAT		VDD_VBAT_RF12	VDD_OSD2		ON_OFF1	VDDP_DIG_MS	M1	M0	M7	M9	MEM_AD[9]	MEM_AD[13]		MEM_CAS_n	MEM_BFCLKO0		MEM_BFCLKO1	Reserved	14
13	Di3_TX_DATX	Di3_RX_DATX		F26M	VDD_VPLL		ON_OFF2	M6	M4	M5	M8	M10	MEM_AD[12]	MEM_AD[14]		MEM_RAS_n	MEM_WR_n		MEM_SDCLKO	Reserved	13
12	F32K	VDD_RTC	VSS_RTC	VDD_LVDS	DPLUS	CC_IO	RESET1_N	PWR_ON_CMP					MMCI1_CLK	MEM_CS0_n	MEM_CS1_n	MEM_CS2_n	MEM_RD_n	VSS_MAIN	Reserved	Reserved	12
11	OSC32K	VSS_MAIN		VSS_MAIN	DMINUS		CC_CLK	ALERT_N		VDD_CORE	VSS_MAIN		MMCI1_CMD	MMCI1_DAT[0]		MEM_CSA0_n	MEM_CKE		Reserved	Reserved	11
10	USB_DMINUS	USB_TUNE		VDD_USB_DIG	CC_RST		REF_CLK_EN	HW_MON1		MMCI1_DAT[2]	MMCI1_DAT[3]		VDDP_MMC	MMCI1_DAT[1]		TRIG_IN	TRST_n		Reserved	Reserved	10
9	USB_DPLUS	VBUS1	VDD_USB	VDD_USB_HSPHY	VDD_CORE	CLK32K	RESET_CMP	HW_MON2					MMCI1_CD	VDD_FUSE_LS	TCK	FCDP_RBn	RESET2_N	FWP	VSS_MAIN	Reserved	9
8	USB_TEST	USB_ID		MMCI2_CMD	CLKOUT0		CMP_EN	I2S1_RX	I2S1_CLK0	VDDP_DIG	TDO	GPIO_035	GPIO_041	VDD_FUSE_FS		VDD_CORE	MIPI1_DATA7		Reserved	Reserved	8
7	GPIO_001	GPIO_000		MMCI2_DAT[0]	VSS_MAIN		VSS_MAIN	I2S1_WA0	I2S1_WA1	VSS_MAIN	TDI	GPIO_036	GPIO_029	VSS_MAIN		VSS_MAIN	MIPI1_DATA6		Reserved	Reserved	7
6	GPIO_002	GPIO_007	NC	MMCI2_DAT[1]	VDDP_DIG				SWIF_TXRX			GPIO_030				VDDP_DIG	MIPI1_DATA5	NC	MIPI1_DATA2	Reserved	6
5	GPIO_003	GPIO_008		MMCI2_DAT[2]	MMCI2_DAT[3]	VDD_CORE	DSP_AUDIO_IN1	I2S1_TX	CHRG_EN	USIF1_RXD_MRST	TMS	GPIO_037	GPIO_031	GPIO_025	VDD_CORE	GPIO_020	MIPI1_DATA4		MIPI1_DATA1	Reserved	5
4	GPIO_004	GPIO_009		MMCI3_CD	MMCI2_CLK	VSS_MAIN	MMCI2_CD	USIF1_TXD_MTSR	USIF1_RTS_N	USIF1_CTS_N	USIF3_SCLK	GPIO_038	GPIO_032	GPIO_042	VSS_MAIN	GPIO_021	MIPI1_DATA3		MIPI1_DATA0	MIPI1_CLK	4
3	GPIO_005	GPIO_010				VDDP_DIG			VSS_MAIN			VSS_MAIN			VDDP_DIG				GPIO_013	GPIO_011	3
2	GPIO_006	MMCI3_CMD	MMCI3_DAT[0]	MMCI3_DAT[2]	T_OUT0	FSYSEN	XRESET	I2C1_SCL	I2C2_SCL	USIF3_TXD_MTSR	USIF3_RTS_N	GPIO_039	GPIO_033	GPIO_027	GPIO_024	CLKOUT2	GPIO_018	GPIO_016	GPIO_014	GPIO_012	2
1	NC	MMCI3_CLK	MMCI3_DAT[1]	MMCI3_DAT[3]	T_OUT1	SYSCLKEN	I2C_INT	I2C1_SDA	I2C2_SDA	USIF3_RXD_MRST	USIF3_CTS_N	GPIO_040	GPIO_034	GPIO_028	GPIO_026	GPIO_023	GPIO_019	GPIO_017	GPIO_015	NC	1
	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	T	U	V	W	Y	

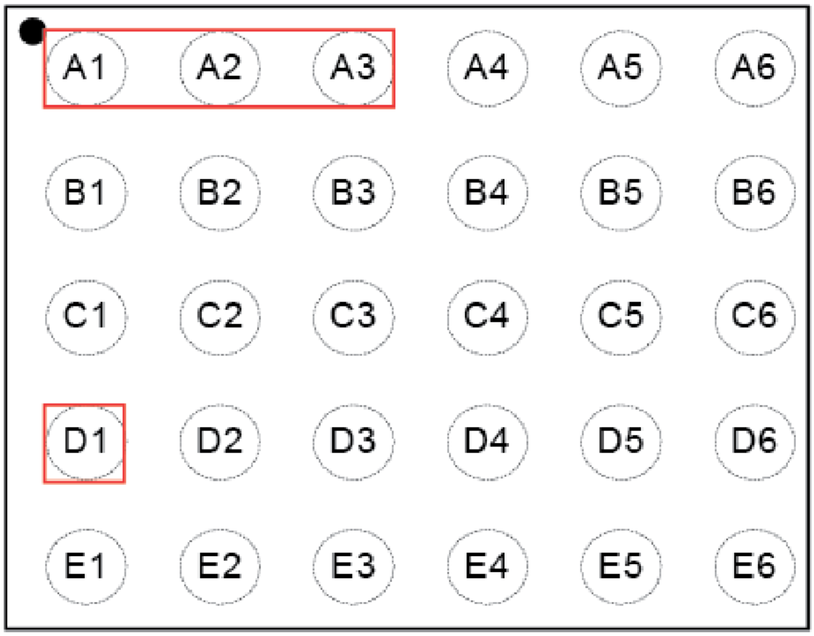
U601

: NC


27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
A
B
C
D
E
F
G
H
J
K
L
M
N
P
R
T
U
V
W
Y
AA
AB
AC
AD
AE
AF
AG

U702

: NC


A1
A2
A3
A4
A5
A6
B1
B2
B3
B4
B5
B6
C1
C2
C3
C4
C5
C6
D1
D2
D3
D4
D5
D6
E1
E2
E3
E4
E5
E6